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Symmetry-Aware Learning in Machine Intelligence: Architectural Principles and Deployment Trade-offs

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Abstract

Symmetry-aware machine learning (ML) embeds invariance and equivariance constraints directly into model architectures, providing principled inductive biases that can improve generalization, sample efficiency, and alignment with data-generating processes. This survey presents a principled synthesis of symmetry-aware architectures, treating symmetry as the core architectural design axis, rather than as an auxiliary modeling property. We introduce a structured taxonomy of symmetry types by mapping discrete, continuous, and statistical group actions to their architectural realizations, including group-equivariant convolutions, equivariant graph networks and attention mechanisms, neural operators, and symmetry-constrained generative models. Beyond architectural categorization, the survey examines how symmetry constraints shape the representation capacity, training stability, robustness, and computational behavior. Representative deployment scenarios are analyzed across molecular modeling, scientific ML, three-dimensional (3D) vision, and relational reasoning, with explicit attention to symmetry-aligned benchmarks, evaluation protocols, performance regimes, inference costs, hardware compatibility, and interactions with low-precision quantization. In addition to conceptual synthesis, the survey consolidates representative empirical indicators reported in the literature, including benchmark outcomes and computational complexity characteristics across major symmetry-aware-architecture families. Across multiple domains, reported studies frequently indicate improved sample efficiency, robustness under symmetry-preserving transformations, and reduced prediction error compared with unconstrained models when symmetry assumptions align with the underlying data-generating processes. In contrast to prior studies that emphasize abstract formalism or isolated domains, this work integrates theoretical foundations, architectural mechanisms, empirical evidence, and system-level considerations into a unified analytical framework, highlighting emerging directions in scalable symmetry-aware learning and hardware-aware co-design.

Keywords: equivariant neural networks, group symmetry, architectural inductive bias, quantization-aware deployment, symmetry-aware model design

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1 Introduction

Symmetry constitutes a foundational design principle in modern ML, particularly in settings where data exhibit invariance or equivariance under structured transformations. Spatial translations, geometric rotations, and index permutations reflect intrinsic constraints that can be systematically embedded into model architectures [1]. By aligning inductive biases with these structural regularities, symmetry-aware models restrict the hypothesis space in a principled manner, leading to improved generalization, enhanced sample efficiency, and more predictable reasoning behavior [2, 3]. For example, molecular properties remain unchanged under rigid rotations and translations of atomic coordinates, object identity in 3D perception is preserved under changes in viewpoint, and graph-level predictions should be invariant to arbitrary node indexing. In such settings, explicitly encoding the corresponding symmetries allows models to generalize across equivalent configurations without repeatedly observing each transformed instance during training.

As ML systems are increasingly deployed in data-scarce, high-dimensional, or physically governed environments, the explicit incorporation of symmetry has become a design necessity rather than a theoretical refinement. Domains such as molecular modeling, scientific computing, 3D perception, and relational reasoning naturally require architectures that respect the underlying transformation structures. At the same time, symmetry enforcement introduces nontrivial challenges related to expressivity, scalability, optimization dynamics, and deployment constraints, highlighting trade-offs that extend beyond model accuracy alone [4]. These opportunities and challenges have motivated a rapidly expanding body of work spanning group-theoretic foundations, neural architectures, scientific ML, geometric deep learning (DL), and large-scale deployment settings.

Despite the rapid progress in symmetry-aware learning, existing research remains largely compartmentalized. Prior studies typically emphasize specific architectural families or formal equivariance properties, often in isolation from system-level considerations and empirical evaluation regimes. Consequently, the relationship between symmetry constraints, architectural design choices, and real-world deployment remains insufficiently synthesized, limiting the translation of theoretical advances into practical and scalable learning systems [5]. Moreover, the literature remains fragmented not only at the architectural level but also in the empirical evaluation of symmetry-aware models, with benchmark results, computational characteristics, and deployment trade-offs typically reported in isolation across various domains. As a result, practitioners lack consolidated reference points for comparing symmetry-aware architectures across tasks, datasets, and computational regimes.

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1.1 Motivation and Key Contributions

The motivation for this survey stems from the observation that symmetry-aware learning, despite its technical maturity and rapid growth, has evolved along largely disjoint theoretical, architectural, and application-driven directions. Consequently, equivariance principles, architectural realizations, and deployment considerations are often examined in isolation, obscuring their interdependencies and system-level consequences. This fragmentation limits a holistic understanding of how symmetry constraints simultaneously affect representation capacity, learning dynamics, scalability, computational costs, and real-world applicability. A unified treatment is therefore required to position symmetry as a coherent design principle rather than a collection of isolated modeling techniques. Such a perspective is increasingly important as symmetry-aware methods are now routinely employed across diverse scientific, geometric, and relational learning settings, where architectural choices, computational requirements, and deployment constraints interact in fundamentally different ways.

This survey addresses this gap by explicitly treating symmetry as a system-level design axis in modern ML. The main contributions of this study are as follows:

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- Integrates mathematical formalism, architectural mechanisms, algorithmic behavior, and deployment considerations into a unified analytical perspective.
- Systematically classifies symmetry types based on underlying group actions and links them to concrete architectural realizations, including group-equivariant convolutions, equivariant attention mechanisms, and neural operator formulations.
- Analyzes the influence of symmetry constraints on representation capacity, training stability, robustness, and computational trade-offs across empirical regimes.
- Reframes symmetry as a primary architectural design paradigm with direct implications for model construction, optimization behavior, and deployment feasibility, rather than as a secondary modeling enhancement.
- Synthesizes representative empirical results and computational characteristics reported in the literature, providing comparative summaries of representative benchmark outcomes, architectural complexity, and deployment trade-offs across major symmetry-aware model families.

1.2 Methodology

This study follows a structured and transparent evidence synthesis methodology aligned with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [6] framework, which was adapted to the requirements of cross-domain surveys in symmetry-aware ML. The methodology is designed to ensure methodological rigor, reproducibility, and comprehensive coverage of the theoretical foundations, architectural mechanisms, and deployment-relevant considerations. No explicit publication date restrictions were imposed. Foundational works are included when they introduce core symmetry concepts or architectural principles that remain central to contemporary models, whereas more recent studies are included when they extend these foundations towards modern architectures, scalable training regimes, or practical deployment contexts.

A comprehensive literature search was conducted across major peer-reviewed bibliographic databases, including IEEE Xplore, ACM Digital Library, SpringerLink, and ScienceDirect. Google Scholar was used as a complementary resource to improve recall across research communities and to capture influential contributions spanning multiple domains. Search queries were formulated using structured Boolean expressions that combined symmetry-oriented terminology with architectural and domain-specific descriptors from the literature. Symmetry-related concepts include equivariance, invariance, group actions, group representations, Lie groups, permutation symmetry, invariant representations, symmetry-aware architectures, and geometric DL. These were combined with architectural terms covering group-equivariant convolutional networks, graph neural networks (GNNs), equivariant message passing, SE(3)- and E(3)-equivariant models, tensor field networks (TFNs), equivariant transformers and attention mechanisms, neural operators, normalizing flows with symmetry constraints, and physics-informed neural networks (PINNs). To capture cross-domain usage and deployment-relevant perspectives, symmetry and architecture terms were further combined with descriptors related to structured and relational data, molecular and physical systems, 3D perception and robotics, scientific ML, multimodal reasoning, robustness under transformations, generalization behavior, computational efficiency, scalability, and hardware-aware considerations. Terms within each conceptual category were linked using OR operators, while intersections across the symmetry, architecture, and domain categories were enforced using AND operators. The query syntax was adapted to the conventions of each database to ensure consistency and comprehensive coverage.

Only peer-reviewed journal articles, magazines, and major conference papers were considered for the primary inclusion in this review. In contrast, technical reports, theses, preprints, and white papers were excluded from the core corpus. The search returned 166 records, of which 148 unique studies remained after the removal of duplicates. The titles and abstracts were screened to

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determine whether each study explicitly addressed symmetry-aware learning at the model or architectural level. Studies limited to informal invariance claims, generic data augmentation, or purely mathematical treatments without architectural or learning implications were excluded at this stage. Full-text screening further assessed whether the remaining studies provided a clear formalization of symmetry, an identifiable architectural mechanism enforcing or approximating equivariance or invariance, or an analysis of expressivity, robustness, computational cost, or deployment-related trade-offs attributable to the symmetry constraints. Studies that did not meet these criteria were excluded from the synthesis. An overview of the literature identification, screening, and selection processes following the PRISMA framework is illustrated in Fig. 1.

The final corpus retained for analysis consists of 104 peer-reviewed references, complemented by 5 additional resources related to benchmark datasets, software frameworks, and widely used implementation toolchains that are directly relevant to the development and evaluation of symmetry-aware models. Each retained study was analyzed using a structured coding process that captured the type of symmetry addressed, architectural mechanism employed, application domain, and nature of the reported evaluation or deployment considerations. This process enabled consistent cross-study comparisons and supported the integrative synthesis presented in this survey.

In addition to qualitative synthesis, the structured coding process enabled the aggregation of representative empirical indicators reported across the literature, including benchmark performance metrics, computational complexity indicators, and architectural resource requirements. These indicators were not treated as standardized experimental evaluations but were synthesized in comparative summaries to facilitate cross-architectural analysis and to provide practitioners with consolidated reference points for understanding the empirical regimes and deployment trade-offs of symmetry-aware models.

The adopted methodology emphasizes breadth, historical continuity, and analytical rigor and is intended to support a representative, conceptually grounded synthesis of symmetry-aware ML that spans several domains.

The survey was organized according to the conceptual structure shown in Fig. 2. This article follows a coherent progression from formal symmetry principles to architectural realizations, comparative analysis, and application-oriented perspectives, treating symmetry as the central design axis throughout. This organization supports a unified examination of how the group-theoretic structure informs model design, empirical behavior, and system-level considerations, before concluding with a synthesis of the open research gaps and future directions.

The remainder of this paper is organized as follows. Section 2 presents a structured taxonomy of symmetry categories in ML literature. Section 3 examines how these symmetries are operationalized using specific architectural mechanisms and learning algorithms. Section 4 provides a comparative assessment of symmetry-aware architectures concerning unconstrained models. Section 5 explores practical deployments across domains, with attention to benchmark designs and task-specific evaluation metrics. Section 6 outlines the emerging research directions and the unresolved challenges. Section 7 interprets the findings of the survey. Finally, Section 8 concludes the paper.

2 Categories of Symmetry in Machine Learning

Symmetry plays a foundational role in modern ML, guiding the design of models that respect structural invariance of data. This section develops a rigorous taxonomy of symmetry by integrating group-theoretic formalism with architectural and algorithmic considerations.

2.1 Definitions and Basic Formalism

The notion of symmetry in ML is rigorously formalized through group theory, where the transformations of inputs or intermediate representations are characterized by the actions of groups. In this

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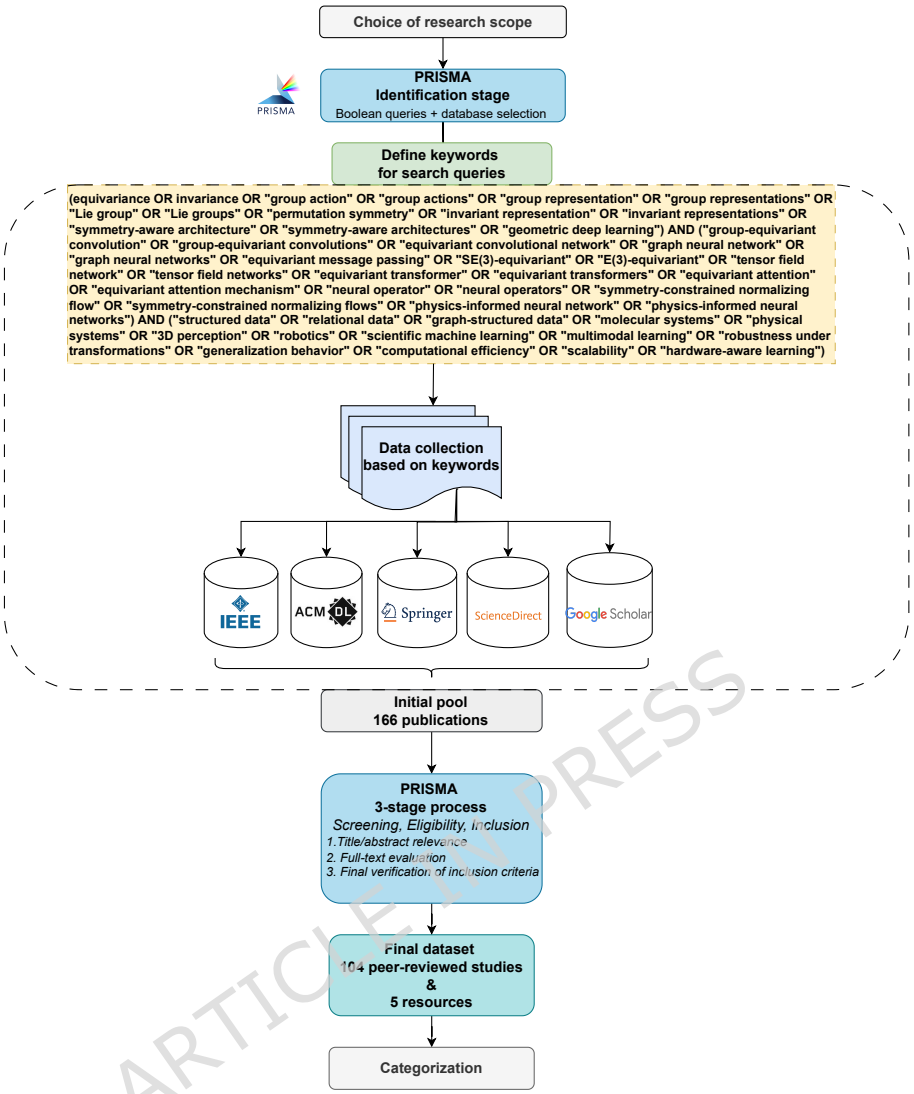


Fig. 1 Overview of the Literature Search Pipeline and Multi-Stage Filtering Process for Survey Inclusion.

framework, symmetries are treated as algebraic constraints that, when embedded in model architectures, induce inductive biases that are aligned with the structure of the data. Such biases promote improved generalization by enforcing invariance or equivariance under transformations that are semantically irrelevant or structurally intrinsic [7]. Let G denote a group and X a non-empty set, typically representing a space of inputs. A group action of G on X is a mapping $\rho : G \times X \to X$, commonly written as $\rho(g, x) = g \cdot x$, satisfying two axioms: (i) for the identity element $e \in G$, it holds that $e \cdot x = x$ for all $x \in X$, and (ii) for all $g_1, g_2 \in G$ and $x \in X$, the compatibility condition $(g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x)$ is satisfied. These properties ensure that the group acts consistently on the input space and that multiple transformations are composed coherently. Given a function $f : X \to Y$, where Y is the output space of the learning task, the symmetry properties of f are defined in relation to its interaction with the group's action. If the output remains invariant under transformations of the input, then f is said to be invariant under G. Formally, this means that $f(g \cdot x) = f(x), \quad$ for all $g \in G$ and $x \in X$. In this case, the output does not change when the input

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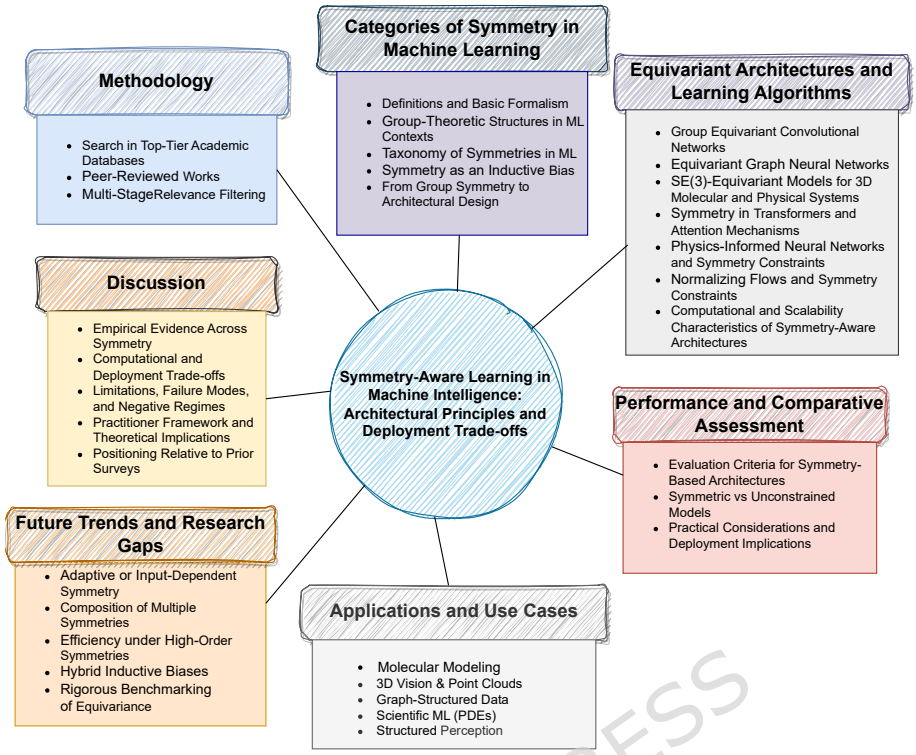


Fig. 2 Conceptual structure and thematic organization of the survey.

is transformed according to a group element, which is essential in classification tasks in which the target label is semantically independent of certain transformations [8].

More generally, if group G also acts on output space Y, and function f preserves the group action between X and Y, then f is called equivariant with respect to G. In this setting, for all $g \in G$ and $x \in X$, the following identity holds: $f(g \cdot x) = g \cdot f(x)$. Equivariance implies that the transformation of an input yields structured and predictable outputs. This concept is central to many modern neural network architectures, especially when outputs such as heatmaps, feature maps, or structured predictions must be transformed consistently with their inputs [9]. A central aspect of invariance and equivariance is that they impose compatibility constraints between the function f and the underlying group action. Such constraints may be encouraged algorithmically through data regularization or augmentation, or enforced architecturally through model design. Although data augmentation can empirically improve robustness to transformations, it does not guarantee strict equivariance at the level of internal representations. In contrast, architectures explicitly constructed to be equivariant, such as group convolutional networks, satisfy these constraints by design and ensure consistent behavior under group actions [10]. These definitions provide the basis for classifying symmetry types according to the nature of group G, its algebraic structure, and its actions on the input and output spaces.

Beyond their formal role, invariance and equivariance have profound implications for learning dynamics, representation efficiency, and model generalization. By constraining the admissible function class to transformations that are consistent with the underlying group action, symmetry-aware models effectively reduce the hypothesis space that must be explored during optimization. This reduction introduces a structured inductive bias that can substantially improve the sample efficiency, particularly in domains where the available observations represent only a small fraction of the possible symmetric configurations of the underlying phenomenon. Equivariance further

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enables systematic parameter sharing across transformed instances of the same pattern, allowing models to reuse the learned representations rather than independently acquiring equivalent features under multiple transformations. Consequently, symmetry-aware architectures frequently achieve improved statistical efficiency, enhanced robustness under symmetry-preserving perturbations, and more stable generalization behavior than unconstrained counterparts when the assumed transformation structure accurately reflects the data-generating process [11, 12].

At the same time, symmetry constraints introduce a fundamental trade-off between inductive bias and representational flexibility. While stronger symmetry assumptions reduce the effective complexity of the learning problem, they also restrict the space of the admissible solutions. Consequently, the benefits of invariance and equivariance are not universal but depend critically on the validity of the underlying symmetry assumptions. In many real-world settings, symmetries are only approximately satisfied owing to measurement noise, boundary effects, partial observability, environmental variability, or latent factors that break the exact group structure. Under such conditions, strict equivariant enforcement may over-constrain the learned representations and limit the predictive performance, despite preserving desirable mathematical properties. This observation has motivated increasing interest in approximate, relaxed, and data-adaptive notions of symmetry, which seek to balance formal equivariance guarantees with the flexibility required by complex real-world learning environments [13].

2.2 Group-Theoretic Structures in ML Contexts

The implementation of symmetry in ML is not merely a matter of conceptual alignment but a direct consequence of the algebraic and topological structure of the transformation group G. Although the general notion of equivariance is defined relative to a group action, the concrete realization of this property depends critically on the class of groups considered, the nature of their representations, and the associated computational frameworks used. A canonical example is the discrete translation group $\mathbb{Z}^n$, which acts on lattice-structured domains that are typical in modern ML applications. Convolutional neural networks (CNNs) are equivariant to this group by construction: for a function $f : \mathbb{Z}^n \to \mathbb{R}$, the discrete convolution with a filter ψ is given by $[f * \psi](x) = \sum_{g \in \mathbb{Z}^n} f(g^{-1} \cdot x)\, \psi(g)$, where the group operation corresponds to the additive translation. Equivariance is guaranteed by the convolution theorem: if τ_g denotes translation by g, then $\tau_g(f * \psi) = (\tau_g f) * \psi$. In this case, the convolution operation intertwines with the group action, making the layer-level transformation G equivariant [14]. Beyond translation, symmetry under discrete permutations arises when the data points are unordered or structurally symmetric. The symmetric group S_n, consisting of all permutations of n elements, acts naturally on input tuples by re-ordering the coordinates. For a function $f : \mathbb{R}^n \to \mathbb{R}$ to be invariant under S_n, it must satisfy $f(x_1, \ldots, x_n) = f(x_{\pi(1)}, \ldots, x_{\pi(n)})$ for all $\pi \in S_n$. Models that operate on sets, such as Deep Sets, enforce this constraint via decomposition theorems, where any continuous S_n-invariant function admits a representation of the form $f(x_1, \ldots, x_n) = \rho\left(\sum_{i=1}^{n} \phi(x_i)\right)$, for suitable functions ϕ and ρ. This structural result illustrates how group-theoretic invariance can be embedded into the network architecture [15].

In continuous settings, groups such as the special orthogonal group $SO(n)$ and the special Euclidean group $SE(n)$ govern the symmetries in geometric and physical data. The group $SO(3)$ comprises all orientation-preserving rotations in 3D space and acts on $\mathbb{R}^3$ via linear transformations that preserve the inner products. The general form of a group convolution on a compact Lie group G is given by $[f * \psi](x) = \int_G f(g^{-1} \cdot x)\, \psi(g)\, d\mu(g)$, where μ is the Haar measure on G. This formulation underlies the equivariant architectures on non-Euclidean domains and requires the integration of harmonic analysis and representation theory into neural network design. The choice of group affects both the space of the allowable filters and the complexity of the convolutional operations [16]. To encode equivariance under $SO(3)$ or $SE(3)$, modern architectures such as SE(3)-Transformers or

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TFNs rely on group representations, that is, homomorphisms $\rho : G \to \mathrm{GL}(V)$, where V is the vector space. These representations describe how the latent features are transformed by the action of G. A map $f : X \to V$ is equivariant if $f(g \cdot x) = \rho(g)f(x)$, ensuring consistency with the group structure, across layers. In equivariant message-passing frameworks, these transformations are applied to the node and edge features, preserving the physical or geometric consistency of the predictions across the graph domain [17].

The choice of symmetry group has implications that extend far beyond mathematical expressivity and directly influence computational efficiency, scalability, and deployment feasibility. Translation-equivariant architectures operating on discrete groups such as $\mathbb{Z}^n$ benefit from highly optimized convolutional implementations and, therefore, incur relatively modest computational overheads. Permutation-invariant and permutation-equivariant models introduce additional aggregation and message-passing operations but remain tractable for large-scale applications. In contrast, equivariance with respect to continuous groups, such as $SO(3)$ and $SE(3)$ typically requires spherical harmonic decompositions, tensor-product operations, or group integration procedures that substantially increase the computational complexity, memory requirements, and implementation difficulty. Consequently, the practical cost of enforcing symmetry generally increases with the algebraic richness of the underlying transformation group. This computational hierarchy partly explains why highly expressive equivariant architectures remain concentrated in scientific ML, molecular modeling, and physical simulation domains, where the additional computational burden is often justified by the strong structural priors encoded in the data [18].

Symmetry in ML does not always arise explicitly from the input space but may emerge implicitly in the latent representations. For example, certain variational models exhibit symmetries under reparameterization, affecting the identifiability and posterior structure, which can be analyzed using group-theoretic constructs such as orbit spaces or gauge structures. From a design perspective, the structure of group G, whether it is finite, compact, or continuous, imposes practical constraints on equivariant modeling. Finite groups permit direct equivariant mappings, whereas continuous or non-compact groups typically require approximations, such as low-order harmonic expansions or discretized group actions. These design choices directly influence the expressivity and computational feasibility of equivariant architectures [19].

More generally, the relationship between symmetry and model expressivity is inherently characterized by a trade-off. Although stronger equivariance constraints embed valuable prior knowledge into the learning process, they simultaneously limit the range of functions that a model can represent. As additional symmetry requirements are imposed, the architecture becomes increasingly specialized towards a particular transformation structure and may lose flexibility when the assumed symmetries are incomplete, approximate, or partially violated in practice. This issue is particularly relevant in real-world environments, where noise, boundary conditions, missing observations, and latent contextual factors frequently break exact group invariance. Under such circumstances, strict equivariance may introduce representational bottlenecks that reduce predictive accuracy, despite preserving mathematically desirable properties. Consequently, a growing body of recent research explores approximate, adaptive, and data-dependent formulations of symmetry that seek to balance the benefits of equivariant inductive biases with the flexibility required to model complex and imperfectly symmetric phenomena [20].

2.3 Taxonomy of Symmetries in ML

The formalism introduced above permits a principled classification of the symmetries relevant to ML systems. We distinguish between three primary forms of symmetry—global, local, and statistical—and two complementary enforcement paradigms, namely approximate and implicit symmetry formulations, which relax or operationalize symmetry constraints in practical learning environments. Together, these categories capture both the mathematical structure of symmetry and the

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mechanisms through which it is incorporated into modern learning systems. *Global symmetries* refer to transformations that act uniformly across the entire input space. Mathematically, these correspond to group actions $g \cdot x$ where the group G acts homogeneously on all components of $x \in X$. Canonical examples include translation and rotation invariance in visual processing tasks. For such symmetries, if $f : X \to Y$ is the target function and $f(g \cdot x) = f(x)$ holds for all $g \in G$, the model can exploit this invariance to reduce the sample complexity. CNNs exemplify this through convolutional layers that are equivariant to discrete translation groups $\mathbb{Z}^n$ [21]. *Local symmetries*, from a taxonomic perspective, act on subsets or substructures of the input space. They may be formalized using sheaf-theoretic or fiber bundle formulations, in which the group action is defined over open covers or neighborhoods of manifolds. In GNNs, for instance, local permutation invariance within node neighborhoods underlies the expressive power of message passing. Formally, for a graph $\mathcal{G} = (V, E)$ and a node $v \in V$, the function f applied to its neighborhood $N(v)$ satisfies $f\left(\{x_u : u \in N(v)\}\right) = f\left(\{x_{\pi(u)} : u \in N(v)\}\right), \quad \forall \pi \in S_{|N(v)|}$, indicating local invariance under the symmetric group $S_{|N(v)|}$. These local symmetries are effective when the input has an inherent, compositional, or relational structure [22].

Statistical symmetries capture invariance at the distributional level rather than the functional level. These arise when the data-generating distribution $\mathbb{P}_X$ is invariant under the action of a group G, that is, for all measurable $A \subseteq X$ and $g \in G$, we have $\mathbb{P}_X(A) = \mathbb{P}_X(g \cdot A)$. This form of symmetry can be exploited by learning models that minimize the risk under distributional shifts aligned with G. In the probably approximately correct (PAC)-Bayesian framework or domain generalization settings, incorporating statistical symmetries may lead to improved generalization across domains related to group transformations [23]. This classification also extends to notions such as *soft* and *approximate symmetries*, which allow for equivariance tolerance. These are relevant when the underlying group action is only approximately valid or when the model architecture does not support a strict equivariance. For instance, suppose the equivariance error is defined as $\epsilon_{\text{eq}}(g, x) = \|f(g \cdot x) - g \cdot f(x)\|_Y$, where Y is the output space with a norm structure. One can enforce soft equivariance by minimizing the expected equivariance error over a group measure μ, $\mathbb{E}_{g\sim\mu, x\sim\mathbb{P}_X}\left[\epsilon_{\text{eq}}(g, x)^2\right]$. This relaxation enables the incorporation of symmetry as a regularization term during the training process [24].

The distinction between strict and approximate symmetry is particularly important in practical learning environments. Although exact equivariance provides strong theoretical guarantees, many real-world data sources exhibit only partial or imperfect symmetry. Measurement noise, sensor calibration errors, occlusions, boundary effects, discretization artifacts, and latent contextual variables frequently violate the exact group structure, even when an approximate symmetry is present at a higher level of abstraction [13]. Similar challenges arise in biological, social, and behavioral systems, where the underlying processes are rarely governed by an exact transformation invariance. Under such conditions, architectures that enforce strict equivariance may inadvertently discard task-relevant information, reduce representational flexibility, and over-constrain the representation space. Consequently, approximate symmetry formulations often provide a more favorable balance between inductive bias and representational flexibility, allowing models to exploit dominant structural regularities while remaining robust to deviations from idealized symmetry assumptions [25].

Symmetry constraints can be enforced explicitly through architectural design or implicitly via data augmentation and regularization. Explicit approaches guarantee equivariance by construction, whereas implicit methods trade formal guarantees for greater flexibility. Unlike soft equivariance, which relaxes symmetry constraints through loss formulation, implicit approaches approximate invariance indirectly via data or training procedures. The symmetry categories introduced above are not mutually exclusive. In practice, modern learning systems often combine multiple symmetry mechanisms and enforce them at different stages in the learning pipeline. Accordingly, the presented taxonomy serves as an analytical framework for organizing symmetry concepts rather than

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Table 1 Taxonomy of Symmetries in ML Architectures.

Symmetry Type	**Group G**	**Group Action $\rho(g, x)$**	**Constraint on f**	**Representative Architectural Manifestations**
Global	$\mathbb{Z}^n$, $SO(n)$	$g \cdot x$ acts uniformly on $x \in X$	$f(g \cdot x) = f(x)$ or $f(g \cdot x) = g \cdot f(x)$	CNNs, Group CNNs, SE(3)-Transformers, TFNs
Local	S_k, local permutation groups	Permutations or transformations on $N(v) \subset X$	$f(\{x_u\}) = f(\{x_{\pi(u)}\})$, $\pi \in S_k$	Message Passing GNNs, Equivariant GNNs, Local Equivariant Modules
Statistical	Any G with $\mathbb{P}_X(g \cdot A) = \mathbb{P}_X(A)$	g acts on the data distribution	Risk invariance under group-induced distribution shifts	Invariant risk minimization, domain generalization
Approximate	Continuous, discrete, or approximate group actions	Relaxed or approximate $\rho(g, x)$	$\mathbb{E}_{g,x}\left[\epsilon_{\text{eq}}(g, x)^2\right]$ minimized	Symmetry-Regularized Models, Relaxed Equivariant Networks
Implicit	Any G (no architectural enforcement)	Simulated via data or priors	No explicit constraint on f; invariance induced through training	Data augmentation, Bayesian priors encoding symmetry

a strict partitioning of methods. Moreover, the coexistence of multiple symmetry types within a single learning system often necessitates hybrid enforcement strategies, combining architectural equivariance, symmetry-aware objectives, and data-driven regularization mechanisms.

Table 1 synthesizes the main symmetry types encountered in ML, linking each category to its underlying group structure, nature of the group action, mathematical constraints imposed on the model, and the corresponding architectural manifestations of the model. This highlights the differences between global, local, statistical, approximate, and implicit symmetries, both algebraically and in terms of their implementation within learning systems.

Although the symmetry categories presented above provide a useful conceptual taxonomy, they also imply distinct modeling trade-offs. Global symmetries often yield substantial gains in sample efficiency and parameter sharing but may become overly restrictive when the transformation assumptions are only approximately valid. Local symmetries provide greater scalability and adaptability to structured data but can introduce limitations associated with localized receptive fields and incomplete structural awareness. Statistical symmetries improve robustness under distributional variation but depend critically on the validity of the assumed invariance properties of the data-generating distribution. Approximate symmetry formulations offer increased flexibility and improved tolerance to real-world deviations from idealized group actions, albeit at the cost of weaker theoretical guarantees. Implicit symmetry mechanisms based on augmentation or regularization are generally easier to deploy but do not ensure consistent equivariant behavior throughout the learned representation hierarchies. Consequently, the selection of an appropriate symmetry formulation is not solely a mathematical consideration but also a design decision shaped by the expected transformation structure, the available data, computational constraints, and deployment requirements. In particular, no single symmetry formulation is universally optimal; the most effective choice depends on the extent to which the assumed transformation structure reflects the true invariances and equivariances of the underlying task.

2.4 Symmetry as an Inductive Bias

The integration of symmetry into learning systems serves not only as a mathematical constraint but also as a fundamental inductive bias that shapes the hypothesis space in a structured and data-efficient manner. Inductive bias refers to the set of assumptions that a model uses to generalize beyond the observed data. When symmetry is incorporated, the learning algorithm implicitly restricts the class of admissible functions to those that are invariant or equivariant under specific

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group actions, thereby reducing the effective complexity of the function class and improving generalization, particularly in low-data regimes [12]. Formally, let $\mathcal{H}$ denote the hypothesis space of functions $f : X \to Y$. Imposing symmetry under a group G leads to the definition of a restricted hypothesis class $\mathcal{H}_G \subset \mathcal{H}$ such that for every $f \in \mathcal{H}_G$, we have either $f(g \cdot x) = f(x)$ (invariance) or $f(g \cdot x) = g \cdot f(x)$ (equivariance), for all $g \in G$. This constraint transforms the learning problem into an optimization over $\mathcal{H}_G$, often resulting in a lower sample complexity owing to the reduced number of degrees of freedom [25]. From a Bayesian perspective, symmetry can be encoded using prior distributions over functions or parameters that favor equivariant solutions. For example, priors over weights in a neural network can be constructed to enforce symmetry using shared parameters or structured covariance. In this view, symmetry becomes an epistemic constraint, reflecting the belief in the structure of the target function based on the domain knowledge [26].

Moreover, the symmetry-based inductive bias naturally aligns with the theory of representation learning. Representations that transform predictably under group actions can capture the essential data structure, thereby enabling compositionality and disentanglement. In equivariant models, the layers are designed such that their intermediate representations are transformed according to a specific group representation. If $\rho_{in} : G \to \mathrm{GL}(V_{in})$ and $\rho_{out} : G \to \mathrm{GL}(V_{out})$ are the group representations acting on the input and output spaces of a layer, then a linear map $\Phi : V_{in} \to V_{out}$ is equivariant if $\Phi(\rho_{in}(g)v) = \rho_{out}(g)\Phi(v), \quad \forall g \in G,\ v \in V_{in}$. These equivariant linear maps are the building blocks of architectures that preserve symmetry across layers, forming the basis of group CNNs and TFNs [27]. Symmetry enhances robustness and generalization under structured distribution shifts by preserving prediction consistency when data transformations align with a group G to which the model is equivariant. This consistency enables parameter sharing across symmetric configurations, thereby improving the data efficiency and transferability. Moreover, explicit symmetry priors provide interpretability benefits because the model behavior under known transformations can be formally analyzed and predicted [23].

The practical value of symmetry as an inductive bias has been repeatedly demonstrated across domains in which the underlying transformation structure is known a priori. In molecular property prediction, equivariant architectures operating on three-dimensional geometric representations have frequently been reported to outperform comparable unconstrained models by exploiting the rotational and translational symmetries inherent to molecular systems [28]. Similar advantages have been observed in physical simulations, protein structure modeling, and three-dimensional vision, where symmetry-aware architectures frequently achieve improved sample efficiency, stronger generalization under symmetry-preserving or geometry-induced distribution shifts, and enhanced robustness to geometric transformations [29]. These reported empirical trends support the theoretical expectation that symmetry constraints reduce the effective search space of admissible solutions and enable learning algorithms to focus on task-relevant variations rather than redundantly relearning equivalent patterns across transformed observations. Consequently, symmetry can be interpreted not only as a mathematical property of a model but also as a mechanism for converting domain knowledge into statistically efficient learning behavior.

Although symmetry provides a principled inductive bias that constrains the hypothesis space, it does not guarantee improved predictive performance. Empirical gains from equivariance arise only when the imposed symmetry accurately reflects the underlying data-generation process [2]. In such settings, approximate or softly enforced equivariance often provides a more appropriate inductive bias than rigid architectural constraints, motivating approaches that interpolate between strict, relaxed, and learned symmetry-enforcement mechanisms. This trade-off highlights a recurring theme throughout the symmetry-aware learning literature: the effectiveness of equivariant inductive biases depends not only on the mathematical validity of the assumed symmetry but also on the extent to which that symmetry remains informative under realistic data conditions and deployment constraints.

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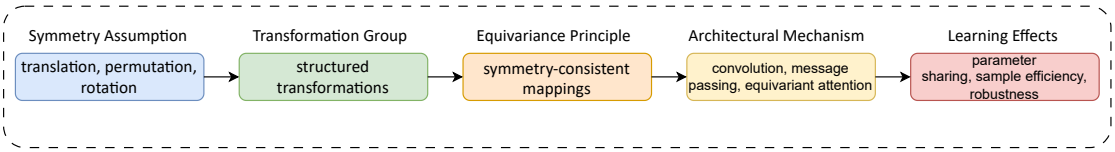


Fig. 3 Conceptual pipeline linking group-theoretic symmetry assumptions to architectural design mechanisms in modern ML models.

2.5 From Group Symmetry to Architectural Design

In practical ML systems, symmetry principles are operationalized by translating group-theoretic constraints into architectural design mechanisms that govern parameter sharing, feature transformations, and information propagation throughout the learning process. While group theory provides a mathematical formalism for describing invariance and equivariance, modern neural architectures implement these constraints through specific parameter-sharing strategies, equivariant operators, or symmetry-preserving aggregation functions.

For example, discrete translation symmetry in grid-structured data leads to convolutional layers that share parameters across spatial locations, ensuring an equivariance to shifts in the input. Similarly, permutation symmetry in unordered sets motivates architectures such as Deep Sets or permutation-equivariant GNNs, where aggregation operators enforce invariance to element reordering. In geometric learning tasks involving 3D structures, continuous symmetry groups, such as $SO(3)$ or $SE(3)$ are incorporated through spherical harmonic representations, TFNs, or equivariant attention mechanisms. These examples illustrate how group actions defined in the mathematical formalism directly inform the architectural constraints that enforce symmetry in practical ML models.

Importantly, the architectural realization of symmetry is not uniform across learning paradigms. Different symmetry groups impose distinct representational, computational, and scalability requirements, leading to substantially different design choices across architectures. Discrete symmetries can often be enforced through relatively simple parameter-sharing schemes, whereas continuous geometric symmetries typically require specialized equivariant operators, harmonic representations, or tensor-valued feature transformations. Consequently, the practical implementation of symmetry involves a balance between the theoretical guarantees, expressive capacity, computational cost, and deployment feasibility. Understanding this relationship is essential for evaluating why different symmetry-aware architectures exhibit distinct performance characteristics and application domains despite sharing a common group-theoretic foundation.

Fig. 3 articulates the conceptual pipeline through which symmetry assumptions translate into architectural design decisions, linking group-theoretic structures to concrete implementation mechanisms in modern ML systems.

3 Equivariant Architectures and Learning Algorithms

Incorporating symmetry into ML architectures introduces principled inductive biases that align the learned representations with the structural invariance in the data, thereby improving generalization and robustness. This section presents a mathematically grounded treatment of convolutional, graph-based, sequence, and flow-based architectures that encode group-theoretic symmetries, covering both strictly equivariant models and approaches that relax equivariance in response to the domain structure, data quality, and deployment constraints.

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3.1 Group Equivariant Convolutional Networks

Classical CNNs exhibit equivariance to discrete translations, which plays a central role in their success with grid-structured data such as images. However, many application domains involve richer symmetry structures, including rotations, reflections, and other transformations beyond translational invariance. To explicitly incorporate such symmetries into the learning process, group-equivariant CNNs (G-CNNs) have been proposed as generalizations of standard CNNs. The key idea of G-CNNs is to replace conventional convolutions with group convolutions that operate on functions defined over symmetry groups. Let G be a group that acts on the input domain, and let $f : G \to \mathbb{R}^C$ represent a feature map defined over the group, where C denotes the number of channels. The group convolution of f with a filter $\psi : G \to \mathbb{R}^C$ is defined by $[f * \psi](g) = \sum_{h \in G} f(h^{-1}g)\, \psi(h)$, where the group action is embedded in the shift of the convolution, and the summation is taken over the elements of G. This operation ensures that the resulting feature maps are transformed consistently within the group, meaning that equivariance is preserved, layer by layer [30].

Formally, equivariance is expressed using the left-regular representation $L_k f(g) = f(k^{-1}g)$ for $k \in G$. If the input is transformed by L_k, then the group convolution satisfies $[L_k f * \psi](g) = L_k[f * \psi](g)$, preserving the symmetry structure through each convolutional layer. This property holds for the entire network, provided that the nonlinearities commute with the group action [31]. To implement G-CNNs in practical architectures, the group G is typically chosen to be finite or discretized. For image data, popular choices include cyclic groups and the planar symmetry group $p4m$, which comprises translations, 90-degree rotations, and reflections. These groups allow convolution over both spatial and orientation dimensions, enabling parameter sharing across transformed feature configurations and improving robustness to symmetry-preserving variations in visual data [32].

Filter steerability is also central to G-CNN design. A steerable filter transforms under group actions as a linear combination of basis filters, allowing weight sharing and compact parameterization. In this context, harmonic networks and spherical CNNs utilize basis filters derived from representation theory and harmonic analysis to achieve computationally tractable group convolutions on a sphere or in 3D space [33]. The practical effectiveness of G-CNNs depends strongly on the extent to which the assumed symmetry group reflects the transformation structure of the data. When rotational or reflectional symmetries are genuinely present, group convolutions can substantially improve the sample efficiency by enabling parameter sharing across multiple transformed versions of the same pattern. This property often reduces the need for extensive data augmentation and improves the robustness of symmetry-preserving transformations [11]. However, these benefits are accompanied by additional computational and memory costs because feature representations must be maintained over the group domain rather than only the spatial domain. Furthermore, when the assumed symmetry is only weakly present or task-relevant information depends on absolute orientation, strict equivariance may restrict representational flexibility and limit the predictive performance. Consequently, the effectiveness of G-CNNs depends not only on their theoretical equivariance guarantees but also on the degree to which the underlying symmetry assumptions accurately characterize the target domain.

3.2 Equivariant GNNs

Permutation symmetry in graph-structured data arises from the invariance of the graph structure to the node relabeling. To respect this symmetry, GNNs are designed to be permutation equivariant at the level of node representations, ensuring that re-ordering node indices induces a corresponding permutation of intermediate features while preserving graph-level invariance [34]. Let $G = (V, E)$ be a graph with node set V and edge set E, and let $\mathbf{h}_v^{(k)} \in \mathbb{R}^d$ denote the feature vector of node v in layer k. A general permutation-equivariant message-passing scheme can be expressed as: $\mathbf{h}_v^{(k)} =$

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$\phi^{(k)}\left(\mathbf{h}_v^{(k-1)}, \square_{u\in\mathcal{N}(v)}\psi^{(k)}\left(\mathbf{h}_u^{(k-1)}, \mathbf{h}_v^{(k-1)}\right)\right)$, where $\mathcal{N}(v)$ denotes the neighborhood of node v, $\psi^{(k)}$ is the message function, $\phi^{(k)}$ is the update function, and $\square$ is a permutation-invariant aggregation operator (e.g., sum, mean, or max). The overall construction ensures equivariance under node permutations; permuting the node indices results in a corresponding permutation of the output features [35]. This equivariant design principle is essential in applications such as molecular property prediction, where atoms correspond to graph nodes, and the model must produce consistent outputs under the re-ordering of atom indices. Architectures such as SchNet, DimeNet [36], and E(n)-equivariant GNNs [37] further specialize this principle into geometric graphs by incorporating additional symmetries such as rotational or translational invariance in 3D space.

The symmetry-aware design of equivariant GNNs provides substantial benefits in relational domains by enabling parameter sharing across graph permutations and reducing dependence on arbitrary node indexing. These properties reduce the sensitivity to arbitrary node orderings and facilitate the transfer of learned relational patterns across structurally similar graphs. However, the locality of message-passing operations introduces important limitations. Information is propagated through successive neighborhood aggregations, which can make it difficult to capture long-range dependencies, higher-order interactions or global structural properties. Furthermore, multiple graph structures may become indistinguishable under certain aggregation schemes, limiting their expressive power despite exact permutation equivariance. These limitations are particularly relevant in scientific and geometric learning tasks, where local neighborhood information may be insufficient to characterize the global topology of the underlying structure. Consequently, recent research has increasingly explored higher-order message-passing schemes, attention-based graph architectures, and hybrid symmetry-aware models that combine equivariant inductive biases with mechanisms capable of capturing richer structural interactions. This evolution highlights that symmetry preservation and representational expressivity must be jointly considered when designing graph-based learning systems.

Closely related symmetry-preserving principles also appear in set-based architectures, such as Deep Sets and Set Transformers, which provide a complementary framework for permutation-invariant learning. These models rely on universal approximation theorems for permutation-invariant functions, which state that any continuous invariant function $f : \mathbb{R}^{n\times d} \to \mathbb{R}$ on a finite set can be decomposed as $f(\{x_1, \ldots, x_n\}) = \rho\left(\sum_{i=1}^{n}\phi(x_i)\right)$, where ϕ and ρ are suitable functions learned by the network. This decomposition is mathematically justified and ensures that the learned function respects the fundamental symmetries of the input domain. Together, equivariant GNNs and set-based architectures extend symmetry-preserving learning principles beyond grid-structured domains to relational, geometric, and unordered data. These architectures incorporate both algebraic symmetry constraints and permutation-invariant aggregation mechanisms, enabling learning systems to remain consistent with the structural symmetries of relational, geometric, and set-structured data [38].

3.3 SE(3)-Equivariant Models for 3D Molecular and Physical Systems

Symmetry in 3D geometric data often arises from the physical constraints governing the domain. In molecular systems, robotics, and physical simulations, the function to be learned must remain consistent under Euclidean transformations, such as translations and rotations. These symmetries are captured by the special Euclidean group SE(3), which combines SO(3) rotations and translations in $\mathbb{R}^3$. Learning functions that are equivariant under SE(3) ensure consistent transformation behavior under changes in the coordinate system, yielding physically meaningful and frame-independent representations [39]. In this context, a model $f : \mathbb{R}^{3n} \to \mathbb{R}^m$ is said to be SE(3)-equivariant if, for any transformation $g \in \mathrm{SE}(3)$ and input configuration $X = (x_1, \ldots, x_n)$, it satisfies: $f(g \cdot X) = \rho(g)f(X)$, where $\rho(g)$ denotes a representation of SE(3) that acts on the

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output space. Equivariance at this level enables models to preserve structural relationships, such as distance and angle, regardless of global orientation [40].

Several architectural paradigms have been developed to realize these equivariance properties in practice. TFNs utilize irreducible representations of SO(3) to construct features that are transformed into spherical tensors. These architectures generalize convolution operations to 3D space by employing spherical harmonics $Y_\ell^m(\theta, \phi)$ and Clebsch-Gordan coefficients for equivariant tensor products. The resulting propagation mechanism guarantees that feature transformations remain consistent under rotations, ensuring that geometric information is preserved across layers while respecting the underlying group structure [18]. While TFNs rely primarily on harmonic representations and tensor algebra, subsequent architectures have explored alternative mechanisms for enforcing equivariance while improving scalability and modeling flexibility.

SE(3)-Transformers adapt attention mechanisms to enforce SE(3) equivariance by constructing queries, keys, and values as vectors or tensors that are consistently transformed under rigid motions. Attention weights are computed using relative positional encodings that preserve rotational symmetry, thereby ensuring spatially consistent interactions [41]. By enforcing consistency under rigid transformations, SE(3)-equivariant models can exploit geometric regularities that would otherwise need to be learned from data alone, yielding representations that remain stable under changes of reference frame and coordinate orientation. However, these benefits come at a substantial computational cost. The use of spherical harmonics, tensor-product operations, and higher-order feature representations significantly increases memory consumption and computational complexity relative to conventional graph or transformer architectures. Furthermore, strict equivariance may become restrictive when geometric symmetries are only approximately satisfied because of the measurement noise, discretization artifacts, conformational variability, or incomplete observations. Consequently, the practical effectiveness of SE(3)-equivariant models depends on balancing geometric inductive biases against computational scalability and the degree to which exact geometric symmetries are satisfied in the target domain [29].

3.4 Symmetry in Transformers and Attention Mechanisms

Transformer architectures are built around attention mechanisms that determine the interactions among the positions within a sequence. Standard attention becomes permutation-sensitive when it is combined with positional encoding or sequence order assumptions. To incorporate symmetry in this setting, architectures must be adapted to enforce invariance or equivariance under input re-ordering or translation [42].

One of the most direct methods for achieving symmetry is to design permutation-invariant attention mechanisms. This requirement naturally arises in set-structured data or in tasks where the ordering of elements has no semantic meaning. Let $X = \{x_1, x_2, \ldots, x_n\}$ be an input set, and consider an attention-based function f that acts on X. For f to be permutation-invariant, it must satisfy $f(X) = f(\pi(X))$ for any $\pi \in S_n$, where $\pi(X)$ denotes the permuted set. Set Transformers address this by using attention modules in which the query vectors are fixed or pooled, ensuring that the output remains invariant to input permutations [43]. While permutation invariance is appropriate for unordered sets, sequence and spatial domains require mechanisms that preserve meaningful positional relationships while remaining compatible with relevant symmetry constraints.

Relative positional encoding provides a symmetry-aware design. Instead of encoding absolute positions, which breaks translation equivariance, relative positional encoding parameterizes attention scores as functions of relative positional offsets rather than absolute indices. Given token positions i and j, the attention mechanism depends on relative offsets such as $(i - j)$ instead of positions i and j independently. Under appropriate formulations, this yields translation-equivariant attention maps and is critical in domains such as speech and protein modeling, where spatial or

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temporal shifts should not alter the model predictions [44]. These ideas naturally extend from sequence modeling to relational domains, where symmetry is defined not by positional offsets but by graph automorphisms and node permutations.

In graph-based transformers, symmetry is enforced using attention mechanisms that are equivariant to the graph automorphisms. Equivariant attention layers ensure that re-ordering the nodes of a graph leads to consistent permutations of output features. Formally, if G is a graph with automorphism group Γ, and f is a transformer layer acting on node features X, then for all $\gamma \in \Gamma$, it holds that $f(\gamma \cdot X) = \gamma \cdot f(X)$. These properties are guaranteed using equivariant message-passing schemes or invariant pooling functions over the neighbors [45].

Attention mechanisms can incorporate symmetry priors by constraining the attention weights or restricting the parameter space to the symmetric functions. In equivariant transformer layers, the linear transformations used for key, query, and value computations are replaced by group-equivariant linear maps, ensuring that the attention scores and outputs are transformed consistently under group actions. This formulation has been extended to geometrically structured domains, such as point clouds and manifolds, where $SE(3)$- or $E(n)$-equivariant transformers enforce spatial consistency [46].

The integration of symmetry into attention mechanisms offers greater flexibility than many earlier equivariant architectures because global interactions can be modeled directly through attention rather than through localized convolution or message-passing operations. This property is particularly advantageous in domains where long-range dependencies and global structural relationships are critical. However, symmetry-aware transformers typically inherit the computational complexity of attention mechanisms while introducing additional constraints associated with equivariant representations. Moreover, the benefits of symmetry-aware attention depend strongly on the alignment between the imposed symmetry assumptions and the underlying structure of the data. When transformation symmetries are weak, approximate, or only partially relevant to the prediction task, strict equivariant constraints may reduce representational flexibility without providing commensurate performance gains. Consequently, current research increasingly explores hybrid transformer architectures that combine symmetry-preserving components with adaptive attention mechanisms capable of accommodating deviations from idealized symmetry assumptions.

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3.5 PINNs and Symmetry Constraints

PINNs integrate domain-specific knowledge into neural models by embedding governing physical laws, typically represented by partial differential equations (PDEs), directly into the learning process. Instead of relying solely on data loss, PINNs incorporate the residuals of PDEs into the loss function, thereby enforcing physical consistency across space and time. This framework is particularly powerful when the available data are sparse or partially observed because it leverages the known physical symmetries to constrain the hypothesis space [47]. Symmetry plays a central role in domains governed by structural constraints, where conservation laws arise as direct consequences of continuous symmetries formalized by Noether's theorem. When a physical system is equivariant under transformations such as time translation, spatial rotation, and reflection, its governing equations contain conserved quantities. Within the PINN framework, such symmetries can be explicitly encoded through loss formulations or architectural constraints, enabling physically consistent learning [48].

Mathematically, suppose that the solution $u(x,t)$ of a PDE transforms consistently under a group action $g \in G$, such that $u(g \cdot x, g \cdot t) = \rho(g)\, u(x,t)$, where $\rho(g)$ denotes a (possibly trivial) representation of G acting on the solution space. The PINN must then be constructed to maintain the equivariance in the network. This can be achieved by (a) incorporating transformed inputs during training (data augmentation over G), (b) using invariant or equivariant basis functions, or (c) parameterizing the network such that its output satisfies the symmetry constraints by design

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[49]. Beyond direct symmetry enforcement within the network architecture, symmetries can also be exploited analytically to reformulate the underlying physical problem itself.

An alternative approach leverages the Lie symmetry analysis to identify the continuous symmetries of a given PDE. These symmetries are then used to reduce the complexity of the learning problem or generate symmetry-adapted coordinates. For example, if a PDE admits a scaling symmetry, the problem can be reparameterized to operate in a scale-invariant form, enabling the PINN to generalize better across multiple regimes [50]. Recent studies have also considered the use of group-equivariant neural operators in PINNs, where the learned solution operator maps function spaces in a manner that commutes with group actions. For instance, an equivariant operator $\mathcal{F}$ satisfies $\mathcal{F}(g \cdot f) = g \cdot \mathcal{F}(f)$ for any input function f and transformation $g \in G$. This perspective unifies PINNs with symmetry-preserving architectures from other domains, ensuring that physical laws and symmetries are preserved not only locally in the solution but also globally in the learned operator [51]. By embedding structural constraints, symmetry-aware PINNs achieve improved data efficiency, physical plausibility and numerical stability. Such formulations have been adopted across fluid dynamics, electromagnetism, elasticity, and quantum mechanics, where respecting the symmetries of the governing equations is often essential for physically meaningful predictions [52].

The effectiveness of symmetry-aware PINNs depends strongly on the validity of the assumed physical symmetries and quality of the governing equations. In practical scientific applications, physical systems often exhibit modeling errors, uncertain boundary conditions, measurement noise, and unmodeled dynamics that violate idealized symmetry assumptions. Under such circumstances, rigid symmetry constraints may introduce optimization difficulties or bias the learned solution towards theoretically consistent but empirically inaccurate behaviors. Moreover, enforcing symmetry through additional loss terms, equivariant operators, or constrained parameterizations increases the training complexity and computational cost. Consequently, recent work increasingly investigates adaptive symmetry enforcement, residual physics constraints, and hybrid operator-learning frameworks that preserve physically meaningful structure without requiring exact adherence to idealized symmetry assumptions.

3.6 Normalizing Flows and Symmetry Constraints

Normalizing flows provide a flexible framework for constructing complex probability distributions through a sequence of invertible and differentiable transformations applied to a simple base distribution, which is typically a standard Gaussian distribution. Let $z_0 \sim p_0(z)$ be a sample from the base distribution, and let the sequence of transformations $f_i : \mathbb{R}^d \to \mathbb{R}^d$, $i = 1, \ldots, K$, define the composite mapping $x = f_K \circ \cdots \circ f_1(z_0)$. The resulting distribution $p(x)$ can be computed using the change-of-variables formula as follows: $p(x) = p_0(z_0) \prod_{i=1}^{K} \left| \det \left(\frac{\partial f_i}{\partial z_{i-1}} \right) \right|^{-1}$, where each f_i must be invertible and its Jacobian determinant is efficiently computable [53].

In the context of symmetry, the design of the transformations f_i can be informed by the known invariance or equivariance properties of the target distribution. For example, if the target density $p(x)$ is invariant under a group action G, that is, $p(g \cdot x) = p(x)$ for all $g \in G$, the normalizing flow must be constructed to respect this symmetry, either by preserving the invariance explicitly or by enforcing the equivariance in conjunction with a symmetry-compatible base distribution [54]. One strategy is to construct flows whose transformation layers commute with the action of the group. If f is equivariant under G, then for all $g \in G$, the following holds $f(g \cdot z) = g \cdot f(z)$. This ensured that the output distribution maintained the symmetry of the input structure. For instance, rotationally invariant flows in 3D molecular modeling applications often leverage spherical harmonics or radial transformations that respect the symmetry of the SE(3) or E(3) group, depending on the modeled invariances [55]. Beyond enforcing symmetry directly through equivariant transformations, alternative formulations seek to exploit symmetry by constructing invariant latent representations.

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Table 2 Comparative Overview of Symmetry-Aware Architectures and Their Symmetry Mechanisms.

Architecture Type	**Target Symmetry**	**Domain Structure**	**Equivariance / Invariance Constraint**	**Core Mechanism**	**Representative Models**
G-CNNs	Discrete or compact groups (e.g., $p4$, $SO(2)$)	Grid-structured inputs (e.g., images)	$f(g \cdot x) = g \cdot f(x)$	Group convolutions, steerable filters	G-CNN, HarmonicNet, Spherical CNN
Equivariant GNNs	Node permutations (S_n)	Graphs, sets	Permutation equivariance of node representations (with graph-level invariance)	Equivariant message passing, invariant aggregation	Graph Isomorphism Network, DimeNet
SE(3)-Equivariant Models	3D rotations and translations	Point clouds, molecules, 3D structures	$f(g \cdot X) = \rho(g) f(X)$	Tensor fields, spherical harmonics, equivariant kernels	SE(3)-Transformer, TFN, E3NN
Transformer Variants	Permutations, translations, and geometric symmetries	Sequences, sets, graphs	Task-dependent invariance or equivariance via architectural bias	Relative positional encoding, equivariant attention, pooling-based attention	Set Transformer, Geometric Graph Transformer, Equivariant Graph Convolutional Layer
Symmetry-Aware PINNs	Continuous Lie group symmetries	PDE-defined physical systems	$u(g \cdot x) = \rho(g)\, u(x)$ or symmetry-constrained residuals	Symmetry-constrained losses, Lie symmetry analysis, equivariant operators	Noether-PINNs, Physics-SymPINN
Equivariant Flows	Density equivariance under G	Latent spaces, manifolds	$f(g \cdot z) = g \cdot f(z)$	Invertible equivariant maps, LieConv layers	LieConv, Equivariant Continuous Normalizing Flow, SO(3)-Flow

Another approach is the use of invariant feature transformations. In this case, rather than modeling $p(x)$ directly, the flow is defined in a latent space, where symmetry-invariant statistics are extracted. This reduces the degrees of freedom and can improve generalization, particularly in structured data settings, such as molecular graphs, rigid bodies, and physical systems governed by conservation laws [56]. More recent developments have incorporated the Lie groups into the normalizing-flow architecture. Flows such as *LieConv* or *Equivariant Continuous Normalizing Flows* define transformations on group manifolds using Lie algebra representations, enabling the model to capture continuous symmetries, such as scaling, translation, or rotation, at a principled level [57].

Symmetry-aware flow models are particularly attractive for physics-informed generative modeling, equivariant density estimation, and uncertainty quantification because symmetry constraints can be incorporated directly into the generative process. By aligning the learned transformations with known invariances or equivariances, these models often achieve improved sample efficiency, robustness, and latent-space interpretability. However, enforcing symmetry within invertible architectures introduces additional design constraints because equivariance must be maintained while preserving the tractable Jacobian computations and exact invertibility. As the complexity of the symmetry group increases, constructing expressive yet computationally efficient equivariant transformations becomes increasingly challenging. Furthermore, when the underlying symmetries are only approximately satisfied, strict symmetry-preserving flows may unnecessarily restrict the expressiveness of the learned density model, preventing the architecture from accurately representing distributional variations that do not conform to the assumed group structure. Consequently, current research increasingly focuses on adaptive and partially equivariant flow formulations that preserve useful symmetry priors while retaining sufficient flexibility to model complex, imperfectly symmetric data distributions.

Table 2 summarizes the principal symmetry-aware architecture families across domains ranging from grids and graphs to manifolds and physical systems, highlighting the associated symmetry groups, architectural mechanisms, and equivariance constraints. Although all of these architectures enforce symmetry-preserving behavior, they differ substantially in representational flexibility, computational requirements, and scalability characteristics, motivating the comparative analysis presented in the following subsection.

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3.7 Computational and Scalability Characteristics of Symmetry-Aware Architectures

Symmetry-aware architectures differ substantially in their computational characteristics because the cost of enforcing equivariance depends directly on the algebraic structure of the underlying symmetry group and the representation mechanisms used for encoding it. Architectures based on discrete symmetries typically rely on parameter sharing and structured convolutional operations, whereas models encoding continuous geometric symmetries often require higher-dimensional representations and specialized equivariant kernels. These design choices lead to distinct trade-offs in terms of parameter efficiency, memory usage, and inference latency across architectural families.

Table 3 notes these computational characteristics for representative symmetry-aware architectures. Models designed for discrete symmetries, such as group-equivariant CNNs and permutation-equivariant GNNs, generally maintain a moderate computational overhead because equivariance can be implemented through efficient parameter sharing and local aggregation operations. In contrast, architectures encoding continuous geometric symmetries, including SE(3)-equivariant networks and related tensor-based models, typically require more expensive representations and therefore incur higher memory and computational costs.

Continuous symmetry groups introduce additional scalability considerations. Table 4 lists representative architectures that encode continuous group actions and the associated complexity of their representation mechanisms. Approaches based on spherical harmonics or tensor-product representations often exhibit increased computational complexity as the representation order grows, whereas coordinate-based equivariant models such as E(n)-equivariant GNNs achieve improved scalability by operating directly on relative positional differences rather than explicit harmonic expansions. This distinction has practical implications for deployment because coordinate-based equivariant models often remain tractable on larger datasets and resource-constrained platforms, whereas harmonic and tensor-based formulations may become prohibitive as model size and representation order increase.

Overall, these comparisons illustrate that the design of symmetry-aware architectures involves balancing theoretical equivariance guarantees with computational efficiency. Although stronger symmetry constraints often improve sample efficiency and generalization when the assumed transformation structure is valid, the associated computational overhead can become substantial for continuous groups and high-order representations. In practical deployments, particularly in resource-constrained environments, the additional memory consumption, inference latency, and implementation complexity of highly equivariant architectures may outweigh their empirical benefits. Consequently, recent research increasingly focuses on coordinate-based equivariant formulations, sparse group representations, low-order harmonic approximations, and hybrid architectures that selectively enforce symmetry only where it provides measurable benefits.

4 Performance and Comparative Assessment

The following section provides a structured assessment of symmetry-aware architectures, relative to unconstrained models. The emphasis is on conceptual trade-offs, generalization behavior, and inductive properties.

4.1 Evaluation Criteria for Symmetry-Based Architectures

Equivariant models induce a restricted hypothesis space $\mathcal{H}_G \subset \mathcal{H}$, which comprises functions that are invariant or equivariant under specific transformation groups. This structural constraint often leads to improved generalization, particularly in low-data regimes, in which the symmetry prior accurately reflects the data-generating process. Such behavior is particularly evident in applications involving geometric or physical symmetries, where enforcing the group structure allows the

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Table 3 Computational characteristics of representative symmetry-aware architecture families.

Architecture Family	Typical Parameter Scaling	Memory Overhead	Inference Complexity
Group-Equivariant CNNs	$\sim \lvert G\rvert^{-1}$ parameter sharing [1, 30]	$\approx$ 1.2–2$\times$ standard CNNs	$\mathcal{O}(HWk^2\lvert G\rvert)$
Equivariant GNNs	$\mathcal{O}(Ed)$ message passing [36, 37]	$\approx$ 1–1.5$\times$ standard GNNs	$\mathcal{O}(Ed)$
SE(3)-Equivariant Models	$\mathcal{O}(Nd^2)$ tensor channels [18, 29]	$\approx$ 3–10$\times$ scalar features	$\mathcal{O}(N^2d)$ attention cost
Transformer Variants with Symmetry Bias	$\mathcal{O}(N^2d)$ attention scaling [44, 46]	$\approx$ 2–4$\times$ standard transformers	$\mathcal{O}(N^2d)$
Symmetry-Aware PINNs	problem dependent [47, 49]	$\approx$ 1.5–3$\times$ baseline PINNs	PDE residual evaluation dominates
Equivariant Normalizing Flows	$\mathcal{O}(d^2)$ invertible transforms [54, 55]	$\approx$ 2–5$\times$ standard flows	Jacobian determinant cost

Table 4 Computational complexity of representation mechanisms for continuous symmetry groups.

Model Family	Symmetry Group	Representation Mechanism	Complexity	Deployment Implication
TFNs	$SO(3)$	Spherical harmonics + tensor products [18, 58]	$\mathcal{O}(L^3)$ in harmonic order L	Expensive for high-order representations
SE(3)-Transformers	$SE(3)$	Equivariant tensor attention [29]	$\mathcal{O}(N^2d)$ attention + $\mathcal{O}(L^3)$ tensor products	Memory-intensive for large-scale systems
Spherical CNNs	$SO(3)$	Harmonic convolution kernels [58]	$\mathcal{O}(L^2 \cdot L^3)$	Moderate scalability limitations
E(n)-Equivariant GNNs	$E(n)$	Relative coordinate differences [37]	$\mathcal{O}(Ed)$	Favorable scalability characteristics
LieConv / Equivariant Flows	Lie groups	Lie algebra parameterization [57]	$\mathcal{O}(d^2)$ transformation cost	Favorable scalability characteristics

model to infer the function behavior over the entire group orbit from a few observations [11]. Enforcing symmetry often introduces additional architectural and algorithmic complexity, as structured convolutional filters, steerable bases, and equivariant attention mechanisms can increase the training cost and inference latency. This overhead may be offset by gains in representational efficiency because symmetry-aware models compactly encode invariant features without redundantly learning equivalent patterns. Consequently, computational efficiency must be evaluated by jointly considering resource consumption, structural parsimony, and generalization benefits [30].

A fundamental trade-off arises between the expressivity and the imposed symmetry constraints. Symmetry-aware models are most effective when the task aligns with the assumed invariances; however, they risk underfitting when the symmetries are only partially valid or when domain heterogeneity violates global equivariance. In such cases, the ability to interpolate between strictly equivariant and more flexible representations becomes critical, motivating the development of hybrid and softly constrained architectures that adaptively modulate symmetry enforcement [24]. In practice, the effectiveness of such constraints depends critically on the degree of alignment between the assumed symmetry group and the true transformation structure of the data, since symmetry mismatch can reduce predictive accuracy despite preserving theoretical equivariance.

By encoding symmetry directly, these models reduce representational redundancy by generalizing across group orbits, rather than learning each transformed instance independently. This compression improves sample efficiency, particularly when transformations correspond to nuisance factors, implying that sample complexity must be evaluated in terms of both the predictive accuracy and data budget required to learn symmetry-induced regularities [59].

The predictable transformation behavior across layers enables symmetry-aware models to learn structured and interpretable internal representations. In architectures that process physical inputs,

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such as $SE(3)$-equivariant networks, intermediate features often carry explicit geometric meanings, facilitating downstream analysis and enhancing model transparency [60]. In such cases, interpretability arises from the alignment of learned representations with domain-intrinsic transformation behavior rather than from post-hoc explanations. When exact equivariance is impractical, approximate or softly enforced symmetry should be evaluated based on stability under structured perturbations, including spatial shifts, node permutations and geometric deformations. In this context, robustness is structural rather than adversarial and reflects the consistency of the model responses to semantically equivalent inputs, with degradation under such perturbations indicating the fidelity of symmetry encoding [61].

4.2 Symmetric vs Unconstrained Models

Symmetry-aware architectures differ from unconstrained models through the explicit integration of structural priors, which define their advantages and disadvantages. These models are most effective in domains where the underlying symmetries are strong and well-characterized, such as physical or geometric tasks, in which enforcing equivariance leads to improved generalization under limited data. In contrast, unconstrained architectures must implicitly infer such structures from data, often requiring larger datasets and remaining more susceptible to overfitting or spurious correlations [12]. By constraining the hypothesis space, symmetric models propagate information across group orbits and avoid degenerate solutions in data-scarce regimes, yielding a more efficient use of the training examples. This effect is particularly pronounced in molecular modeling, 3D perception, and dynamical systems, where symmetries such as translation, rotation, or permutation directly reflect domain semantics or physical laws [28].

Empirical evidence reported across symmetry-aware learning studies indicates that the comparative advantage of symmetry-constrained models is strongly regime-dependent. Generalization gains are most pronounced in low-data settings, where enforcing equivariance enables extrapolation across group orbits from limited observations [62]. As the dataset size increases or when extensive data augmentation is available, the performance gap between the symmetry-aware and unconstrained models typically narrows, indicating that the dominant empirical benefit of symmetry manifests through improved sample efficiency rather than asymptotic accuracy.

Beyond predictive accuracy, empirical evaluations commonly report enhanced stability under transformations that respect the assumed symmetry groups. This form of robustness degrades when perturbations violate the symmetry prior, underscoring the dependence of empirical gains on the fidelity of the imposed structural constraints. Attributing observed performance improvements to symmetry constraints is typically supported by ablation-style comparisons, in which equivariant components are evaluated against otherwise identical unconstrained counterparts, isolating the contribution of the symmetry prior to the architectural capacity. Conversely, empirical failure regimes demonstrate that when symmetries are weak, partial, or input-dependent, strict equivariance can lead to underfitting and reduced adaptability, with unconstrained models achieving superior performance [25].

Unconstrained models retain full representational flexibility and are therefore better suited to domains in which relevant symmetries are unknown, approximate, or vary across the sample sets. In settings such as natural language processing, recommendation systems, or vision tasks with cluttered and heterogeneous data, imposing strict group-based constraints can lead to underfitting or loss of expressivity, with unconstrained architectures often outperforming symmetric models when the symmetry prior is misaligned [63]. Symmetry-aware models may also exhibit brittleness when their foundational assumptions are violated, as enforcing exact equivariance under partially respected symmetries can introduce systematic bias or representational rigidity into the model. Consequently, the choice between symmetric and unconstrained architectures reflects a fundamental trade-off between inductive bias and adaptive flexibility. Symmetry-aware models excel when

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the domain structure is well-defined and reliable, whereas unconstrained models prevail when the structural assumptions are latent, inconsistent, or unknown. As a result, the empirical value of symmetry-aware learning is inherently context dependent, reflecting a balance between the benefits of structural prior knowledge and the flexibility required to model complex real-world phenomena [64].

4.3 Practical Considerations and Deployment Implications

The deployment of symmetry-aware models in real-world systems introduces architectural and operational constraints that differ substantially from standard pipelines. Enforcing equivariance or invariance requires careful parameterization, constraining layer composition, and weight-sharing schemes to preserve group compatibility. For example, group-equivariant convolutional layers must adhere to kernel structures dictated by representation theory, limiting the use of unconstrained filters or positional embeddings. These requirements affect both the model expressivity and design modularity, as architectural components must be composed in a manner that consistently respects the assumed symmetry across transformations [31].

From a computational perspective, the overhead introduced by symmetry-aware architectures strongly depends on the nature of the underlying symmetry group. Models based on discrete and low-order groups typically incur modest additional costs relative to standard architectures, as constrained parameterization and weight sharing reduce representational redundancy without substantially increasing the operator complexity [1]. In contrast, architectures built on continuous symmetries or higher-order group representations often introduce superlinear computational and memory overheads owing to basis expansions, tensor product operations, and group indexing [58]. Consequently, the scalability of symmetry-aware models is closely coupled to both the algebraic complexity of the symmetry group and the order of the representations employed. In practical training pipelines, this increased operator complexity typically results in longer training times and higher peak memory consumption for continuous-group and higher-order equivariant models, whereas architectures based on discrete or low-order symmetries incur substantially smaller overheads, often dominated by standard backbone computations.

Training stability constitutes another dimension in which the symmetry-aware architectures exhibit distinct behaviors. Owing to their reduced hypothesis space and structural alignment, such models often converge more reliably in low-data regimes. However, this advantage can be offset by the numerical instabilities introduced by complex operations, including non-commutative group convolutions, harmonic basis projections, and equivariant tensor contractions [65]. These operations frequently require specialized libraries or custom backends to maintain numerical accuracy, particularly for continuous or high-dimensional groups, thereby increasing the engineering complexity. From an implementation standpoint, equivariant architectures impose stricter structural constraints than conventional models, thereby limiting architectural flexibility and complicating the reuse of pre-trained components. Hardware compatibility further constrains deployment, as operations such as spherical harmonics or steerable convolutions may not be efficiently supported on edge devices or low-power accelerators [66].

Despite these constraints, symmetry-aware models provide robustness and stability under distribution shifts aligned with the assumed symmetry group, making them well-suited for deployment in scientific, medical, and safety-critical applications. Their predictable behavior under structured transformations facilitates debugging and post-hoc verification, as failure modes can often be attributed to violations of group assumptions rather than stochastic noise. However, this rigidity may reduce adaptability when data distributions evolve or deviate from the imposed symmetries. In such settings, hybrid architectures that interpolate between symmetry-aware and unconstrained components can offer more favorable robustness–flexibility trade-offs [20, 67].

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Therefore, the practical value of enforcing symmetry constraints must be assessed against the associated computational overheads. When the assumed symmetry closely matches the data-generating process, the gains in sample efficiency, stability, and physically consistent generalization often justify increased inference costs. Conversely, in large-scale regimes with abundant data, extensive augmentation, or weakly expressed symmetries, the empirical advantage of equivariant architectures frequently becomes marginal, rendering their additional computational burden unjustifiable [12]. Although symmetry-aware models may achieve competitive accuracy with fewer learnable parameters, the representational overhead from group indexing, basis expansions, and tensor-type tracking can increase memory footprint and inference latency. Consequently, effective deployment requires careful engineering to balance theoretical compactness with practical constraints on latency, energy consumption, and memory budget [37]. This trade-off is particularly pronounced for architectures based on continuous symmetry groups, where the computational cost of exact equivariance can dominate the overall deployment budget.

Recent advances in symmetry-aware toolchains have partially mitigated some of the aforementioned deployment challenges. Libraries such as e3nn [68], LieTorch [69], and TFNs [70] provide optimized implementations of equivariant operations, including steerable convolutions and tensor product layers, with graphics processing unit (GPU) acceleration enabled using custom compute unified device architecture (CUDA) kernels [71]. These toolkits are integrated with mainstream DL frameworks, such as PyTorch and JAX, thereby lowering the barrier to incorporating group-theoretic modules into practical workflows. Moreover, Triton-based backends enable low-level kernel optimization for SE(3)- and SO(3)-equivariant architectures, thereby improving the throughput of modern accelerators [72, 73].

Although existing toolchains significantly improve the performance of high-end GPUs, their applicability to edge or low-power inference targets remains limited. Many equivariant operations rely on specialized kernels and floating-point precision, which are inefficiently supported on embedded accelerators, restricting their deployment in latency- and energy-critical environments. Thus, hardware availability and inference constraints play a decisive role in determining whether symmetry-aware models can be practically adopted beyond high-performance computing settings [74].

To address these limitations, several practical strategies have emerged to improve deployment feasibility in resource-constrained environments. These include the use of low-order equivariant representations, coordinate-based formulations that avoid expensive harmonic expansions, approximate equivariance mechanisms, and hybrid architectures that restrict symmetry enforcement to selected network components [75]. Such approaches seek to preserve the most important symmetry-induced benefits while reducing memory consumption, computational complexity, and hardware specialization requirements, thereby improving suitability for edge and low-power deployment scenarios.

Ultimately, the viability of symmetry-aware architectures depends on balancing gains in sample efficiency, robustness, interpretability, and physically consistent generalization against increased rigidity, training complexity, computational overhead, and system integration effort.

5 Applications and Use Cases

Symmetry-aware models are particularly effective in domains where structural invariance is intrinsic to the data and governing processes. Their deployment is most appropriate when group-theoretic priors, such as permutation, rotational, or Euclidean equivariance, align with physical, relational, or geometric consistency requirements. In molecular modeling, for example, predicting atomistic properties under $SE(3)$ transformations requires architectures that are invariant to global translation and rotation while remaining sensitive to interatomic distances and angles [76]. Architectures such as TFNs and SE(3)-transformers have demonstrated robust performance on datasets such

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as QM9, MD17, and OpenCatalyst [77], where metrics such as the mean absolute error (MAE) and root mean squared error (RMSE) serve as standard evaluative criteria for property regression tasks. These datasets are specifically designed to expose the rotational equivariance or invariance behavior of models, thereby enabling fine-grained performance comparisons in chemically relevant areas [78]. A recurring observation in molecular learning benchmarks is that equivariant architectures achieve their largest advantages in data-efficient molecular property prediction and force-field learning, where physically meaningful geometric constraints reduce prediction error and improve generalization across molecular configurations [28].

In 3D vision and robotics, symmetry arises from the object orientation and spatial transformations in point clouds or mesh data. Benchmarks like ModelNet40 [79], and ScanObjectNN [80] provide diverse object categories subjected to random rotations and partial occlusions, making them well-suited for evaluating $SO(3)$-equivariant models such as spherical CNNs or equivariant point cloud transformers. The evaluation metrics vary depending on the task specifications, typically involving classification accuracy or pose estimation errors under rotational perturbations. Equivariance not only improves performance but also enhances robustness to changes in the viewpoint and incomplete geometric data, both of which are common in real-world robotic perception. Empirical studies on these benchmarks consistently indicate that rotationally equivariant architectures retain performance more effectively under viewpoint changes and geometric perturbations than conventional point-cloud models that rely primarily on data augmentation [81].

In the context of graph-structured data, particularly in chemical property prediction and relational reasoning tasks, permutation symmetry governs the node indexing. Equivariant GNNs are evaluated using benchmarks such as ZINC [82], MoleculeNet [83], PubChem Quantum Mechanics 4M Molecules–Large-Scale Challenge (PCQM4M-LSC) [84], and the Open Graph Benchmark (OGB) [85], where node or graph-level targets must remain consistent under graph isomorphism. The ability of the models to maintain prediction consistency across permutations was assessed qualitatively via permutation consistency checks and quantitatively using task-appropriate metrics, such as regression error or classification accuracy, depending on the benchmarks. The GNN architectures employed in such tasks often include message-passing mechanisms that are provably permutation-equivariant, thus ensuring that the model outputs respect the graph symmetries [86]. Reported results across graph benchmarks suggest that permutation-equivariant architectures provide the greatest benefit when relational structure is strongly informative, although the performance advantage often decreases as dataset size and model capacity increase [87].

Scientific ML is another domain in which symmetry-based architectures are particularly effective for achieving high accuracies. PINNs and neural operators are frequently deployed to solve PDEs that exhibit spatial or temporal symmetries. Fluid dynamics, quantum mechanics, and elasticity involve systems governed by conservation laws, scale invariance, and continuous group actions. Benchmarks derived from canonical PDEs like Navier–Stokes, Schrödinger, and Darcy flow provide the testing grounds for such models, with evaluation metrics including L^2 error norms, PDE residual losses, and functional output consistency across symmetry-transformed inputs. The incorporation of symmetry in these contexts leads to increased physical plausibility, improved extrapolation across spatiotemporal regimes, and enhanced generalization under constrained data availability [51]. Across PDE-learning benchmarks, symmetry-aware formulations are most effective when conservation laws and geometric constraints closely match the governing dynamics, yielding improved extrapolation and physically consistent predictions beyond the observed training regime [88].

In structured perception and object-centric scene understanding, symmetry principles emerge as permutations invariance over the input sets or compositional structures. Tasks such as visual question answering, object counting, and logical reasoning in synthetic environments, including compositional language and elementary visual reasoning (CLEVR), require models to reason using unordered sets of inputs. Here, permutation-invariant or equivariant attention-based architectures

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are evaluated for their ability to generate consistent outputs under arbitrary permutations of object indices. Set-based benchmarks emphasize alignment with the symmetry of the task and highlight improvements in the consistency, generalization, and interpretability of relational reasoning processes [89]. Similarly, PointNet-based benchmarks quantify the robustness of spatial permutations in 3D object recognition and scene comprehension [90]. Beyond set-level invariance, recent advances in rotated object detection have demonstrated how geometric symmetry constraints can be explicitly and implicitly enforced at the representation level. In particular, the work in [91] shows that box-level equivariance mechanisms enable accurate orientation-aware localization under weak supervision, while controlled ablations disentangle the contribution of symmetry constraints from architectural capacity. Empirical evidence from structured perception tasks suggests that the primary benefit of symmetry-aware architectures lies in maintaining relational consistency under object or point re-ordering, enabling more reliable compositional reasoning and reducing sensitivity to arbitrary input organization [38].

A consistent conclusion across the application domains surveyed is that the benefits of symmetry-aware architectures are highly dependent on whether evaluation protocols explicitly probe the transformation structures encoded by the model. Metrics that assess robustness, representation stability, or equivariant consistency must explicitly probe the group actions assumed by the architecture; otherwise, the apparent performance gains may reflect symmetry mismatches rather than genuine generalization. Interpretability constitutes an additional evaluative dimension, as explicitly equivariant models expose predictable transformation behavior in intermediate representations, enabling diagnostic analysis and verification in safety-critical settings [92]. In practice, symmetry-aware learning reflects a co-design paradigm rather than a purely architectural choice, requiring joint specifications of symmetry assumptions, model structures, and evaluation protocols. Table 5 outlines this alignment by relating symmetry types to representative application settings, reported empirical signals, and deployment considerations.

6 Future Trends and Research Gaps

Symmetry-aware learning is increasingly influenced by the recognition that real-world data rarely satisfy idealized global invariance. Consequently, recent work has shifted towards adaptive, approximate, and data-dependent symmetry formulations, in which group structures may vary across spatial, temporal, or relational contexts or be partially inferred from data [93]. This trend motivates the development of architectures that dynamically modulate symmetry constraints, thereby enabling selective inductive bias while preserving representational flexibility. Such mechanisms delineate the regimes in which strict equivariance remains effective, as enforcing fixed symmetry assumptions in their absence can lead to systematic generalization failures under realistic symmetry violations.

A fundamental unresolved challenge is determining whether an observed transformation regularity reflects a genuine property of the underlying data-generating process or merely a finite-sample artifact. Current approaches for learning symmetries directly from data remain vulnerable to spurious symmetry discovery, which can introduce unstable inductive biases and degrade out-of-distribution generalization when the inferred symmetry structure does not persist beyond the training distribution [94].

Beyond adaptive formulations, a central research direction is the principled composition of multiple, interacting symmetries. While most existing architectures encode a single transformation group per module, real-world systems often exhibit nested or hybrid symmetries, such as the joint presence of spatial and temporal invariance or the coexistence of permutations and geometric structures in multi-agent settings. Addressing this complexity requires modular and factorized designs that support symmetry composition without sacrificing equivariance, scalability, or differentiability

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Table 5 Synthesis of symmetry types, architectural enforcement strategies, and reported empirical signals in symmetry-aware learning, as observed in several application domains.

Application domain	Symmetry category/type	Symmetry role/enforcement	Model	Benchmark (dataset)	Accuracy/sample-efficiency signal	Generalization/ deployment signal
Molecular Modeling	Global geometric $SE(3)$	Exact equivariance (hard constraint)	Allegro	QM9; MD17/rMD17	Reduced energy/-force error	Out-of-domain (OOD) generalization; scalable large-system molecular dynamics
		Relaxed equivariance (soft bias)	ViSNet	MD17; rMD17; MD22	Lower energy/force error	Transfer to realistic molecules; geometry-aware efficiency trade-off
		Pre-trained equivariant prior	EquiformerV2	OC20; catalytic datasets	Accuracy retained in zero-/few-shot regimes	Cross-domain deployment with substantial speedup relative to density functional theory (DFT)
3D Vision/ Point Clouds	Geometric symmetry (rotations)	Controlled symmetry violations (robustness stress test)	Standard point cloud models (PointNet, DGCNN)	ModelNet40-C	Severe accuracy degradation under geometric corruptions	Limited robustness and poor OOD generalization of non-equivariant models
	Local geometric equivariance	Neighborhood-level consistency	PointNeXt + Local Features	ModelNet40; ScanObjectNN	Improved real-world accuracy with modest computational overhead	Robust transfer across realistic point clouds
	Global geometric $SE(3)$	Exact equivariance (group convolutions)	E2PN	ModelNet40; ScanObjectNN	Accuracy retained with lower computational cost	Stable generalization under rotations
Graph Learning	Permutation symmetry	Controlled symmetry breaking	PG-GNN	Synthetic graph isomorphism; graph classification	Higher expressivity beyond standard message-passing GNNs	Improved performance on structure-sensitive tasks
		Exact permutation equivariance	SMP	Synthetic tasks; ZINC	Improved accuracy on structural tasks	Invariant to node reordering; inductive to unseen graph sizes
		System-level implementation preserving GNN semantics	Edge-conditioned GNNs (system-level DenseScatter)	PCQM4M-LSC; OGB-PPA	Accuracy comparable to OGB baselines	Scalable multi-GPU deployment with preserved equivariance
Scientific ML (PDE Learning)	Geometric, continuous symmetry	Discretization-invariant operators	Neural operators; PINNs	Navier–Stokes; Darcy; Burgers	Resolution-independent predictive accuracy; data-efficient learning	Generalization across meshes and spatiotemporal regimes
Structured Perception	Permutation symmetry (object sets)	Permutation-invariant attention	Object-centric attention models	CLEVR; synthetic scenes	Consistent relational accuracy in low-data regimes	Robust scene understanding under object re-ordering
	Permutation symmetry (point sets)	Invariant feature aggregation	PointNet-style models	ModelNet40	Stable predictive accuracy under permutations	Robust generalization under noise and spatial disorder

[95]. In the absence of such compositional mechanisms, symmetry-aware models remain confined to narrowly structured regimes, limiting their applicability to complex, multiscale systems.

An additional bottleneck is the absence of scalable mechanisms for combining multiple continuous and discrete symmetry groups within a single architecture. Existing formulations often require separate equivariant operators for each group, leading to rapidly increasing computational complexity and making exact compositional equivariance difficult to maintain in realistic large-scale systems [96].

Efficiency under symmetry constraints remains a persistent bottleneck, particularly for architectures that operate on continuous groups and high-order representations. Existing approximations, such as discretized group actions, low-rank projections, and soft equivariance penalties, provide partial relief; however, their fidelity–efficiency–scalability trade-offs remain insufficiently characterized, particularly in large-scale or latency-critical deployments [97, 98]. This limitation is especially

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Table 6 Open Research Gaps in Symmetry-Aware Learning: Bottlenecks and Emerging Directions.

Research Gap	Formalization Gap	Scalability Gap	Evaluation Gap	Key Bottleneck / Emerging Direction
Adaptive / data-dependent symmetry	✓	✗	✗	Spurious symmetry discovery Learning symmetry directly from data
Composition of multiple symmetries	✓	✓	✗	Lack of scalable compositional equivariance across interacting groups / Modular equivariant architectures
Efficiency under high-order equivariance	✗	✓	✗	Representation-cost explosion from tensor products and harmonic expansions / Approximate equivariant representations
Hybrid inductive biases	✓	✗	✗	Conflicting structural priors between symmetry, causality, and compositionality / Unified structural inductive frameworks
Rigorous evaluation of equivariance	✗	✗	✓	Benchmark confounding and lack of controlled equivariance assessment / Symmetry-aware benchmarking protocols

Notation: ✓ denotes a dominant bottleneck; ✗ denotes a secondary or non-dominant factor.

pronounced in architectures relying on spherical harmonics, tensor products, or high-order irreducible representations, where memory consumption and computational cost increase rapidly with representation order. As a result, many theoretically attractive equivariant models remain difficult to deploy in resource-constrained environments despite their favorable learning characteristics [28].

Beyond symmetry in isolation, increasing attention has focused on integrating equivariance with complementary structural inductive biases, including causality, compositionality, and physical consistency. These priors extend beyond group-theoretic constraints and may interact with equivariance in nontrivial ways, for instance, when coordinate equivariance must coexist with conservation laws or modular causal structure. The development of unified frameworks that reconcile symmetry with broader structural principles remains largely unexplored [99, 100].

A key open question is how multiple structural priors should interact when their inductive assumptions are not fully compatible. For example, enforcing strict geometric equivariance may conflict with causal abstractions, task-specific modularity, or partially observed dynamics, creating trade-offs that remain poorly understood from both theoretical and empirical perspectives.

The evaluation of symmetry-aware models remains a challenge. Although equivariance is often demonstrated qualitatively or inferred indirectly from performance gains, few benchmarks explicitly measure symmetry fidelity, robustness under structured perturbations, or deviations from exact equivariance. Consequently, controlled evaluation protocols that isolate the contribution of symmetry constraints are required to enable reproducible and fine-grained comparisons [101, 102]. This gap between the theoretical guarantees of equivariance and realistic symmetry violations underscores the need for principled evaluation methodologies.

Furthermore, many existing benchmarks conflate the benefits of symmetry constraints with improvements arising from larger architectures, additional parameters, or domain-specific engineering choices. Consequently, the field still lacks standardized evaluation protocols capable of isolating the true contribution of equivariance and quantifying the trade-off between symmetry preservation, computational cost, and predictive performance [103]. Table 6 consolidates the identified research gaps by linking them to their dominant methodological bottlenecks and emerging research directions. The table highlights unresolved challenges related to symmetry discovery, compositional equivariance, computational scalability, interactions among structural inductive priors, and rigorous evaluation, thereby providing a consolidated view of the principal barriers to the broader adoption of symmetry-aware learning.

7 Discussion

This section synthesizes the theoretical foundations, empirical observations, and system-level considerations developed throughout this study to extract integrated insights into symmetry-aware

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learning. Rather than reiterating architectural constructions or benchmark-specific results, the discussion distills cross-cutting empirical patterns, trade-offs, and design implications that inform both practical deployment and conceptual understanding of the topic.

7.1 Empirical Evidence Across Symmetry Regimes

While the earlier sections establish the theoretical motivations and evaluation criteria for symmetry-aware learning, this subsection synthesizes recurring empirical patterns reported across comparative studies at a qualitative level. Rather than reiterating individual benchmark results or detailed numerical comparisons, we contrast the empirical regimes under which different forms of symmetry enforcement have been observed to be beneficial or detrimental across the literature.

Across domains, strict equivariance often provides the strongest empirical benefits in data-scarce regimes when symmetry assumptions are well aligned with the underlying data-generating process. As summarized in Table 5, molecular benchmarks such as QM9 and MD17 consistently report reduced energy and force prediction errors for equivariant architectures, graph-learning benchmarks such as ZINC demonstrate improved structural generalization in permutation-preserving tasks, and geometric recognition datasets such as ModelNet40 indicate improved robustness and prediction consistency under rotational transformations. These findings suggest that symmetry-aware architectures achieve their largest empirical gains when physically meaningful or geometrically valid transformation structures are explicitly incorporated into the model design. In such settings, comparative evaluations frequently report improved sample efficiency, robustness, and consistency under symmetry-preserving transformations compared with unconstrained architectures. The empirical signals synthesized in Table 5 further indicate that these gains are most consistently observed in terms of data efficiency, transformation robustness, and physically consistent prediction, whereas improvements in asymptotic predictive accuracy are often more modest and strongly dependent on the validity of the imposed symmetry assumptions. This observation reinforces the view that the primary empirical value of symmetry-aware learning lies in improved inductive bias and sample efficiency, rather than in universally superior predictive performance. As the dataset size increases or extensive data augmentation is employed, the marginal empirical advantage of strict symmetry enforcement is often observed to diminish, with unconstrained or implicitly regularized models achieving comparable performance.

In regimes characterized by noisy observations, partial symmetry violations, or heterogeneous data distributions, approximate or relaxed symmetry mechanisms are more frequently reported to offer favorable trade-offs between inductive bias and representational flexibility. Conversely, empirical studies consistently indicate that misaligned or invalid symmetry assumptions can lead to underfitting or degraded performance, highlighting the sensitivity of symmetry-aware models to violations of their underlying assumptions. Collectively, these empirical patterns reinforce the view that symmetry-aware learning should be evaluated and selected at the level of symmetry regimes rather than individual architectures, with effectiveness determined jointly by the data regime, noise characteristics, and the validity of symmetry assumptions.

7.2 Computational and Deployment Trade-offs

Symmetry-aware learning introduces structural constraints that fundamentally reshape the computational profiles of learning systems. Beyond representational aspects, enforcing symmetry directly affects the training cost, inference latency, memory footprint, and hardware compatibility constraints. This subsection synthesizes the system-level trade-offs associated with different symmetry classes by abstracting individual architectures to highlight the recurring computational patterns reported in the literature. As depicted in Table 7, explicit symmetry enforcement should therefore be treated as a system-level design decision rather than a purely architectural choice. In regimes

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Table 7 System-level training and deployment trade-offs across symmetry classes.

Symmetry Class	Typical Computational Scaling	Training Overhead	Inference Overhead	Deployment Implications
Permutation invariance/equivariance	$\mathcal{O}(Ed)$ message passing [34, 37]	$\approx$ 1–1.3× baseline models	Minimal overhead	Favorable scalability; broad hardware support; suitable for large-scale and edge deployments
Finite discrete group equivariance	$\mathcal{O}(\lvert G\rvert)$ filter replication [1, 30, 32]	$\approx$ 1.5–3× convolution cost	Moderate latency increase	Efficient on GPUs with moderate kernel specialization; manageable deployment complexity
Continuous spatial equivariance ($SO(3)$, $SE(3)$)	$\mathcal{O}(L^3)$ harmonic expansions / $\mathcal{O}(N^2 d)$ attention [18, 29, 58, 76, 97]	3–10× training overhead	High memory and compute cost	Dependent on specialized kernels and high-precision arithmetic; limited suitability for latency-critical or resource-constrained settings
Continuous symmetry in scientific ML	PDE residual evaluation dominates [51, 52, 88]	High training cost	Moderate inference cost	Training-intensive but inference-efficient; well-suited for offline simulation surrogates and scientific pipelines
Latent or learned symmetry mechanisms	Similar to standard deep models [13, 24, 94]	$\approx$ 1–2× overhead	Moderate overhead	Flexible deployment profile; overhead comparable to unconstrained deep models

where physical consistency, robustness, or data efficiency are central requirements, the associated computational overhead is typically justified in reported deployments by the advantages of data efficiency, robustness, and physical consistency. Conversely, in large-scale or latency-sensitive settings, the incremental benefit of architectural symmetry may not outweigh its cost, favoring unconstrained or implicitly regularized alternatives that are aligned with the available hardware and resource budgets.

7.3 Limitations, Failure Modes, and Negative Regimes

Despite their demonstrated benefits, symmetry-aware architectures are not universally advantageous. Their effectiveness depends critically on the alignment between the imposed symmetry constraints and underlying data-generating processes. When this alignment is weak, partial, or violated, symmetry enforcement can restrict the expressivity and induce systematic underfitting, particularly when task-relevant variations are not symmetry-preserving. A central limitation arises from the trade-off between the strict equivariance and representational flexibility. Exact equivariance constrains the hypothesis space in a principled manner but may suppress the informative asymmetries introduced by noise, boundary effects or heterogeneous interactions. In such settings, approximate or relaxed symmetry mechanisms often yield more stable empirical behavior, albeit with weaker formal guarantees, reinforcing the view of symmetry as a tunable inductive bias rather than an absolute architectural requirement.

Empirical evidence further indicates that in data-rich regimes, particularly for tasks requiring invariance, extensive data augmentation and increased model capacity can substantially reduce the marginal performance gains of explicit symmetry constraints, whereas their associated modeling and computational overheads persist, allowing unconstrained architectures to achieve comparable performance at a lower cost [10]. From a theoretical standpoint, equivariance is a strong inductive bias that constrains the hypothesis space of a model, imposing incorrect or overly strict group assumptions that can limit expressivity and induce systematic underfitting when symmetry assumptions mismatch the underlying data structure [5].

Practical feasibility further constrains the adoption of higher-order and continuous-symmetry groups. Models defined over complex Lie groups often incur substantial memory, training, and inference costs, thereby limiting their applicability in latency-sensitive and resource-constrained environments. When symmetry assumptions are only approximately satisfied, this computational burden can outweigh the corresponding empirical advantages. Moreover, real-world deviations

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from idealized symmetry conditions, including measurement noise, discretization artifacts, incomplete observations, and distribution shifts, can degrade the reliability of equivariant guarantees. In deployment-oriented or safety-critical settings, failure to account for such violations may result in brittle behavior, underscoring the need for empirical validation and sensitivity analysis before their adoption. Collectively, these limitations emphasize that symmetry-aware learning is most effective when symmetry assumptions are explicitly validated, context-dependent, and continuously re-evaluated throughout the modeling pipeline, rather than being treated as universally beneficial design principles.

7.4 Practitioner Framework and Theoretical Implications

The empirical evidence, computational trade-offs, and failure regimes synthesized in the preceding sections collectively indicate that symmetry-aware learning should be treated as a conditional design choice, rather than a universally optimal architectural principle. For practitioners, the central question is therefore not whether symmetry should be imposed, but under which conditions explicit symmetry enforcement yields benefits that justify its structural, computational, and deployment costs.

Strict equivariance is most appropriate when domain symmetries are well-defined, task-relevant variations respect these symmetries, and data availability is limited. In such regimes, symmetry constraints effectively reduce the hypothesis space, leading to improved sample efficiency, robustness, and interpretability through principled inductive bias. These conditions are characteristic of physically grounded systems and relational domains governed by stable transformation groups, in which structural assumptions play a decisive role in the generalization. In contrast, when symmetry assumptions are only partially valid, noisy, or locally violated, enforcing exact equivariance can suppress these informative asymmetries and limit representational flexibility. In these settings, approximate or relaxed symmetry mechanisms provide a more effective compromise, offering a stabilizing inductive bias while preserving expressive capacity. Consequently, practitioners should prioritize the empirical validation of symmetry assumptions through sensitivity analysis and ablation-based diagnostics rather than relying exclusively on formal equivariance guarantees.

In data-rich regimes, unconstrained or implicitly regularized architectures often remain competitive, as extensive augmentation, large-scale pre-training, or self-supervised objectives can induce relevant invariances directly from the data. Under these conditions, the marginal gains afforded by explicit architectural symmetry frequently diminish, rendering simpler models preferable from a performance–complexity perspective. The emergence of attention-based architectures, diffusion models, and large-scale foundation models further reinforces this trend, as these systems frequently encode invariance implicitly through training objectives, architectural priors, or scale rather than through explicit group-theoretic constraints. As a result, explicit equivariance should be introduced selectively only when empirical evidence demonstrates benefits beyond those already induced by data availability and training scales. Beyond immediate deployment considerations, these observations have broader theoretical implications. The variability of symmetry-induced gains across data regimes and system constraints underscores that equivariance is best understood as a tunable, context-dependent inductive bias, the effectiveness of which depends jointly on group structure, noise characteristics, data scale, and computational resources. This perspective challenges purely structural accounts of symmetry and motivates theoretical frameworks that integrate symmetry with robustness, approximation, and efficiency considerations.

Overall, this decision-oriented synthesis emphasizes that the value of symmetry-aware learning lies not in universal architectural prescriptions but in the informed alignment of symmetry assumptions with the data characteristics and deployment objectives. Treating symmetry as an adjustable design dimension enables the principled integration of theoretical rigor with empirical performance and practical feasibility in modern ML systems.

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7.5 Positioning Relative to Prior Surveys

The recent proliferation of surveys on symmetry in ML underscores the recognition of structure-aware inductive biases across model classes and application domains. Table 8 provides a comparative synthesis of representative surveys, highlighting the differences in thematic focus, architectural scope, symmetry treatment, and degree of system-level integration. Most existing studies specialize in particular modalities or symmetry groups, offering depth within narrow scopes. In contrast, this survey adopts a unifying analytical perspective in which symmetry is examined as a design abstraction that links representational structure, learning behavior, and system-level considerations across architectural families, with application settings serving primarily as empirical deployment contexts.

Early surveys laid important foundations but predated the formalization of symmetry as an explicit architectural design principle. For instance, [104] provides a comprehensive overview of DL techniques for 3D computer vision, focusing on data representations such as voxels, point clouds, and multiview projections. While highly influential, this work treats invariance largely implicitly through data representations and augmentation rather than through group-theoretic equivariance embedded at the architectural level. Similarly, [2] introduced a unifying vision of geometric DL beyond Euclidean domains, framing symmetry as an organizing concept for learning the graphs and manifolds. However, its emphasis is primarily conceptual and methodological, preceding the emergence of modern SE(3)-equivariant architectures, equivariant attention mechanisms, neural operators, and deployment-aware design considerations. In the same vein, [105] presents a foundational treatment of geometric DL; it predates key developments such as SE(3)-transformers, equivariant neural operators, and recent symmetry-aware hardware-deployment pipelines.

Compared to [106], which explores symmetry primarily as a conceptual bridge between artificial and biological intelligence, this study remains grounded in the computational and algorithmic mechanisms of equivariant learning. The survey in [107] provides a comprehensive treatment of rotation-invariant methods in 3D perception but limits its scope to $SO(3)$-related settings. In contrast, our taxonomy encompasses global, local, statistical, approximate, and latent symmetries applicable across diverse modalities, while explicitly analyzing how these symmetry assumptions shape the representation capacity, training behavior, and system-level trade-offs. Moreover, studies such as [108], focus on geometric GNNs with an emphasis on permutation equivariance and message-passing formulations, while offering limited discussion of system-level considerations such as training dynamics, inference cost, and deployment feasibility. Similarly, [75] emphasizes group-theoretic representations and formal mechanisms of equivariance but provides a limited empirical or hardware-aware analysis. Finally, [109] surveys geometric priors more broadly, treating symmetry as one inductive bias among many rather than as a primary architectural design axis.

Beyond taxonomic breadth, the central contribution of this survey is a cross-cutting synthesis of symmetry as a design principle that jointly shapes the hypothesis spaces, learning dynamics, empirical behavior, and deployment feasibility. This perspective integrates theoretical foundations with empirical regimes and system-level constraints, positioning equivariance not as a purely mathematical abstraction but as a controllable inductive bias, the benefits and limitations of which depend on data regimes, noise characteristics, and computational contexts.

8 Conclusion

This survey positions symmetry not as a peripheral modeling constraint but as a central design axis in modern ML systems. By systematically synthesizing symmetry types, architectural realizations, empirical regimes, and system-level trade-offs, we have shown how equivariance and invariance shape hypothesis spaces, learning dynamics, and deployment feasibility in ways that extend well beyond accuracy-centric evaluation. The analysis indicates that the benefits of symmetry-aware

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Table 8 Comparative summary of representative surveys on symmetry and geometric structure in ML.

Survey	Scope	Model Coverage	Symmetry Types	Deployment Aspects	Benchmark Discussion
[2]	Broad	CNNs, Graphs, Manifolds	Conceptual / Geometric	✗	✗
[104]	Narrow (3D)	CNNs, Multiview, Voxels	Implicit geometric (data-driven)	✗	✓
[105]	Broad	CNNs, GNNs, Manifolds	Geometric (Euclidean, Riemannian)	✗	Partial
[106]	Conceptual	Conceptual only	High-level analogies	✗	✗
[107]	Narrow (3D)	CNNs, PointNets	Rotation ($SO(3)$)	✗	✓
[108]	Narrow (Graphs)	GNNs	Permutation only	✗	✓
[75]	Medium	Group CNNs, Representations	Global (Lie, discrete)	✗	✗
[109]	Medium	CNNs, GNNs, Transformers	Rotation, spatial	✗	Partial
This work	Broad	CNNs, GNNs, Neural Operators, Transformers	Global, Local, Permutation, Approximate, Statistical	✓	✓

Notation: ✓ indicates explicit discussion, ✗ indicates absence, and *Partial* denotes limited or non-systematic coverage. Deployment aspects include training dynamics, computational cost, scalability, and hardware considerations.

architectures are inherently conditional, emerging most strongly when symmetry assumptions align with the data-generating process and diminishing when the data scale, noise, heterogeneity, or task complexity compensate for or violate these constraints. Hence, treating symmetry as an adjustable inductive bias rather than a binary architectural property is essential for understanding when and why symmetry-aware models succeed or fail.

Looking forward, the role of symmetry in ML is likely to evolve from static architectural enforcement to adaptive, compositional, and system-aware integrations. Emerging directions include the combination of multiple interacting symmetries, reconciliation of equivariance with large-scale foundation models and generative processes, and development of mechanisms for discovering, validating, and relaxing symmetry assumptions directly from data. Progress in these directions will require moving beyond idealized symmetry formulations towards frameworks that explicitly account for uncertainty, approximation, and imperfect symmetry structures that characterize real-world environments. Simultaneously, future advances will depend on resolving fundamental challenges related to scalability, compositional equivariance, benchmark design, and the interaction of symmetry with complementary structural priors, such as causality and physical consistency.

Ultimately, the central challenge is not simply enforcing symmetry more rigorously but determining when symmetry should be imposed, relaxed, or adaptively learned. In this context, symmetry-aware learning represents not only a collection of architectural techniques but also a principled framework for co-designing representations, algorithms, and systems. By balancing theoretical guarantees with empirical evidence and deployment constraints, future symmetry-aware methods have the potential to deliver ML systems that are simultaneously more data-efficient, robust, interpretable, and computationally practical across a broad range of scientific and engineering domains.

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Declarations

Conflict of interests. The authors declare that there are no conflicts of interest in this study.

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